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Machining condition variation decoupling adversarial domain adaptation network: A multi-target domain tool wear state recognition method

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ABSTRACT

Accurate tool wear monitoring (TWM) is crucial for enhancing parts quality and production efficiency. Practical metal cutting often necessitates frequent adjustments to working conditions, which introduces data distribution shifts highly correlated with the operating environment when classifying tool wear states across datasets. These lead to domain inconsistency and distribution mismatches, severely impeding multi-target domain adaptation. To overcome the above limitations, this paper proposes a novel machining condition decoupling adversarial domain adaptation network (MCDADAN) for multi-target domain tool wear condition recognition under varying working conditions. First, prototype discrepancy alignment is implemented through a weight-sharing wavelet convolutional feature extractor, which enhances the class prototypes discriminability and ensures clear representation boundaries in the feature space. Second, the adaptive domain discriminator weight is leveraged alongside extracted multi-domain discrepancies to decouple the effects of varying operation conditions. This mechanism effectively suppresses environment-correlated interference features while fully preserving intrinsic tool wear-related features. Moreover, the blended multi-domain alignment framework minimizes source domain and multiple target domains distributional differences and address mismatches encountered during the transfer process. The effectiveness of MCDADAN was validated on two milling datasets (a self-constructed GH4169 dataset and PHM2010 dataset). Experimental results demonstrate that MCDADAN generalizes effectively from a single source domain to multiple target domains, significantly outperforming conventional methods in TWM. The findings indicate its potential for supporting efficient tool replacement decisions during practical machining of difficult-to-machine materials.

1. Introduction

Tool condition monitoring is a key enabler of intelligent manufacturing, supporting higher product quality and machining efficiency while reducing human intervention and improving process reliability through more adaptive control strategies [1]. Previous studies report that tool failures may account for approximately 15%–40% of total production costs and nearly 20% of machine downtime, underscoring their substantial economic impact. In this context, inaccurate tool wear recognition, as a core function of tool

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condition monitoring, can cause direct economic loss and safety risks [2]. Overly conservative decisions may trigger premature tool changes and increased downtime, whereas delayed replacement can lead to scrap parts, degraded surface quality, and even tool breakage or machine damage [3]. Moreover, if a model cannot maintain performance under changing machining conditions, online monitoring becomes unreliable and its industrial deployment is hindered. Therefore, developing robust tool wear monitoring techniques is essential to extend tool life, improve machining quality, and enhance productivity and economic efficiency.

Current tool wear monitoring approaches are conventionally classified as direct and indirect. In direct monitoring, optical or machine-vision systems are used to capture the cutting-edge geometry evolution [4]. However, such methods necessitate machine halt and are often severely impeded by cutting fluids, thereby significantly limiting their applicability in industrial environments. Conversely, indirect monitoring infers tool wear states from in-process sensor data collected during cutting operations [5,6]. Given that these methods exert minimal interference with machining operation and enable continuous operation, they are widely adopted in industrial practice for their more time-efficiency and more suitability for real-time monitoring. Among indirect monitoring techniques, data-driven monitoring has gained prominence due to its capacity for automated analysis of tool condition. During machining, physical signals closely associated with tool wear including cutting force [7], vibration [8], acoustic emission (AE) [9], and machining noise [10], are collected and subsequently analyzed using signal analysis algorithms and machine learning models to indirectly determine the tool wear condition. With the expansion of industrial data ecosystems and the maturation of machine learning, artificial intelligence (AI)-based decision-making models have attracted increasing attention [11]. Typical shallow machine learning models applied in this context include artificial neural networks (ANN) [12], support vector machines (SVM) [13], hidden markov models (HMM) [14], which are used to construct predictive models for tool condition. For example, Kong et al. [15] extracted time-domain, frequency-domain, and wavelet-packet-domain features from milling force signals, fused the resulting descriptors via neighborhood preserving embedding, and employed a whale optimization algorithm-support vector machine (WOA-SVM) model to predict tool wear. Similarly, Meng et al. [16] acquired cutting force, vibration, and acoustic emission signals, derived features in the time, frequency, and time-frequency domains, constructed a sensitive feature matrix, and optimized a back propagation neural network (BPNN) using a gravitational search algorithm. Despite recent advances, conventional machine learning methods still exhibit notable limitations. These approaches place substantial reliance on handcrafted characteristics, which are sensitive to variations in processing conditions and prone to information loss during extraction. Moreover, the limited depth hampers their ability to model the intricate nonlinear dependencies between sensor data and tool wear conditions.

As data volumes continue to expand and computing hardware advances rapidly, deep learning methods have extensively applied in tool wear monitoring, fault diagnosis, and prediction [17–19]. Essentially, deep learning refers to a class of neural-network models with deep architectures, feedforward and backpropagation mechanisms of traditional neural networks. Prominent models used in tool wear monitoring include recurrent neural networks (RNN) [20], convolutional neural networks (CNN) [21], long short-term memory (LSTM) networks [22,23], and their hybrid deep learning architectures. These developments have established two dominant modeling pipelines in the literature. First, models operate directly on raw one-dimensional (1D) multi-sensor signals and learn temporal dependencies end-to-end. Compared to traditional machine learning, the core advantages of deep learning lie in its ability to adaptively extract relevant features from raw signals, and to effectively learn and model the intricate nonlinear mappings between signals and wear states. For instance, Zhang et al. [24] used LSTM networks to model time-series vibration signals acquired during machining, enabling tool wear estimation and remaining useful life (RUL) prediction. Guo et al. [25] proposed a multi-scale pyramid attention network based on LSTM to enhance model hierarchical feature interpretability. A parallel research line recasts signals as time-frequency images to exploit high-capacity two-dimensional architectures. In this pipeline, non-stationary signals are converted into time-frequency representations, (e.g., via the short-time fourier transform (STFT)), and subsequently used to train 2D CNN, residual networks (ResNet), or CNN-Transformer hybrids. Duan et al. [26] applied STFT to convert raw non-stationary signals into time-frequency images and then fed into a CNN model based on ResNet for tool wear state recognition. Martínez-Arellano et al. [27] encoded and reconstructs raw sensor signals as three-channel images, achieving over 90% classification accuracy using CNN. Shu et al. [28] proposed a self-supervised contrastive learning approach built upon CNN, which enabled accurate recognition of tool wear states. Duan et al. [29] further improved noise-robust tool wear prediction from time-frequency images by proposing a multi-scale stacked sparse PCA network with hybrid pooling and sparsity optimization. Dong et al. [30] introduced a VIT-KAN hybrid network, which encoded 1D wear signals into MTF images and enabled parallel feature learning and long-range dependency capture for robust tool wear classification. Li et al. [31] introduced a channel-space attention-based model leveraging wavelet-transformed cutting-force time-frequency images for tool wear state recognition, which substantially reduces misclassification across the fuzzy boundaries between wear stages. Comparative studies consistently report higher accuracy for deep-learning models than for traditional baselines. Aghazadeh et al. [32] directly demonstrated the superior performance of deep-learning approaches to tool wear recognition by comparing CNNs with conventional machine learning approaches, including support vector machines, Bayesian networks and k-nearest neighbors. Complementary to the above pipelines, hybrid frameworks combine multi-sensor fusion with unsupervised or sparse feature learning to enhance robustness and interpretability. Qin et al. [33] developed a hybrid identification and prediction framework combining stacked sparse autoencoders with backpropagation neural networks, achieving accurate monitoring through multi-sensor signal fusion and K-means wear stage labeling. Despite the above progress, substantial challenges persist, particularly the degradation of performance under changing machining conditions and the cost of retraining on the shop floor. Although deep learning has demonstrated strong performance in tool condition monitoring, it still faces limitations in practical application. Specifically, when machining conditions change, the model often exhibits poor adaptability to the new conditions, leading to significant degradation in prediction accuracy. Moreover, retraining models for new machining conditions is time-consuming, severely impacting production efficiency. Therefore, improving the generalization capability and adaptability of deep learning models under varying conditions remains a key challenge in tool wear monitoring.

To address the poor generalization of deep learning models under varying cutting parameters and operating conditions, transfer learning has been introduced for tool wear monitoring. By transferring learned representations from pre-trained models to novel machining scenarios, transfer learning approaches have significantly advanced the development and deployment of tool condition monitoring. Transfer learning enhances model generalization under limited target-domain data by exploiting source-domain knowledge [34], and concurrently reduces the dependence on large-scale labeled datasets, making it effective for cross-domain tool wear monitoring under heterogeneous machining conditions with sparse annotations. In practice, transfer learning has found extensive application in condition monitoring, notably in bearing and gear fault diagnosis [35,36]. Li et al. [37] proposed a domain-adversarial transfer learning framework for bearing fault diagnosis that employs source–target adversarial training to improve predictive performance under new operating scenarios. In gear fault diagnosis, transfer learning has become a mainstream solution for cross-condition monitoring. These successes provide a solid foundation for its application in tool wear identification. Within tool wear monitoring, transfer learning is commonly implemented through model-based and feature-based strategies. Early studies mainly focused on single-target domain scenarios, where training and evaluation were restricted to a single source-domain condition paired with a single target-domain condition. Sun et al. [38] proposed a deep transfer-learning network built upon sparse autoencoders for estimating the RUL of cutting tools, and demonstrated its effectiveness in improving prediction accuracy under limited target domain data. Building on this, Mamledesai et al. [39] developed a transfer learning–based tool condition monitoring method using convolutional neural network, enabling cross-scenario RUL prediction through adaptive feature learning. Further advancing the field, Huang et al. [40] introduced a deep adversarial domain confusion network for cross-domain tool wear monitoring, in which adversarial learning and maximum mean discrepancy (MMD) are jointly employed to align source domain and target domain features and improve generalization capability. To balance the competing objectives of discriminative accuracy and domain alignment in adversarial adaptation, Liu et al. [41] introduced a domain-adversarial neural network whose training relies on multi-loss collaborative optimization to jointly balance discrimination and domain alignment, effectively resolving conflicts between classification and domain alignment losses. Liu et al. [42] developed a domain-separation prediction network equipped with multi-scale spatial-feature visualization to predict milling cutter remaining useful life under different working conditions by enhancing spatiotemporal feature expression and domain adaptability. However, the majority of existing transfer learning methods are designed for single-target domain scenarios, which limits their applicability in multi-condition environments. To improve practical relevance, recent studies have moved toward multi-objective adaptation. Tang et al. [43] designed a proposed network achieves global and local alignment of the source with multiple target domains within hyperbolic space, which effectively mitigates transfer bias. Their method achieved superior fault classification performance under complex cross-domain conditions involving noise and class imbalance. Ragab et al. [44] introduced an adversarial multi-target domain adaptation (MTDA) approach that extends domain-invariant representations learned from a single source domain to multiple target domains. Through adversarial transfer learning, their method achieved outstanding generalization and fault diagnosis accuracy across different bearing datasets. Zhou et al. [45] designed a progressive decoupling adaptation strategy, where target domains are aligned sequentially based on their transfer difficulty. By decoupling domain-specific and domain-invariant features, their method maintains high classification accuracy even under significant distribution shifts. Cai et al. [46] proposed an unsupervised adaptation multi-source domain framework that transforms monitoring data under various cutting parameters into multiple source domains, and effectively aligns their feature distributions with a target domain through multi-feature space adaptation, thereby improving wear state recognition without requiring labeled target data. Kim et al. [47] proposed a mixture density multi-domain network that integrates adversarial learning with Bayesian modeling enabling accurate tool wear prediction across machining conditions with significant variability. Wang et al. [48] developed a dual-level adversarial transfer framework that models task relationships and domain shifts simultaneously, achieving robust remaining useful life estimation across diverse tool configurations and working scenarios.

Multi-target domain adaptation has been actively studied in application areas where a single model is expected to generalize from one labeled source domain to multiple unlabeled target domains, especially in semantic segmentation and related visual recognition tasks. Representative paradigms include collaborative consistency learning with online knowledge distillation, where multiple target-specific experts are trained and then distilled into a single domain-generic student model to perform well across targets. Another line of research formulates MTDA from an information-theoretic perspective by learning a shared latent space common to all domains while modeling domain-private factors to account for target-specific variations. To handle heterogeneous target shifts, progressive MTDA strategies have been proposed to incorporate target domains in an easy-to-hard manner and gradually augment the source with high-confidence target samples, often coupled with representation-decoupled learning. Graph-based MTDA frameworks further exploit cross-domain relationships through semantic propagation or attention mechanisms to facilitate multiple-target feature transfer. MTDA has also been extended beyond segmentation to detection, for example incremental MTDA schemes that adapt to targets sequentially while mitigating forgetting via distillation. In industrial condition monitoring, MTDA has also been explored for rotating machinery diagnostics, where multi-domain representation learning and cross-domain generalization techniques are developed to handle variable operating conditions. However, directly applying generic MTDA pipelines to tool wear monitoring remains challenging. Many existing single-source multi-target methods still prioritize learning a shared feature extractor and enforcing global distribution alignment across domains. This practice may become less robust when target domains exhibit heterogeneous shifts, because aligning only global distributions can overlook category-level mismatches, blur class boundaries, and induce category confusion. Moreover, when adaptation is driven mainly by labeled source supervision, training may become overly source-dependent, resulting in a source-biased classifier. Importantly, in tool wear monitoring, variations in cutting parameters such as spindle speed, feed rate, and depth of cut can substantially reshape signal spectra and may be statistically entangled with wear-sensitive patterns. This condition–wear entanglement makes purely global alignment less reliable, since the learned representation can be biased toward condition-dependent cues rather than the intrinsic tool wear state, thereby limiting the reliability of online monitoring across heterogeneous cutting regimes.

Although multi-target domain adaptation has been extensively investigated in rotating machinery diagnostics, its use in tool wear monitoring remains comparatively limited. Therefore, this paper proposes a novel method to monitor the multi-target domain tool wear state based on machining condition variation decoupling adversarial domain adaptation network (MCDADAN), which is validated using PHM2010 dataset and self-developed GH4169 milling dataset.

- (1) A novel multi-target domain adaptation framework is proposed for tool wear recognition under varying machining conditions, which incorporates adversarial learning and working condition decoupling to enhance cross-domain generalization. To facilitate accurate monitoring, signals—including force, vibration, acoustic emission, and sound—are collected under diverse operating conditions to provide comprehensive monitoring information.
- (2) A prototype discrepancy alignment strategy is introduced to improve the discriminative ability for tool wear stage classification in the representation space by enhancing the separability between source and target domain prototypes.
- (3) A multi-domain blended alignment framework is devised to lessen cross-domain feature divergence among multiple targets domain and alleviate mismatching during transfer.
- (4) A machining condition variation decoupling module is implemented by leveraging domain-discriminator weighting and multi-domain discrepancies to eliminate condition-related interference while preserving intrinsic tool wear features.

The remainder of the paper is organized as follows. Section 2 outlines the theoretical background. Section 3 provides a comprehensive description of the MCDADAN methodology. Section 4 presents the validation of the proposed approach using GH4169 milling experiments and the PHM2010 dataset. Section 5 offers the conclusions.

2. Theoretical foundation

2.1. Classic domain adversarial neural network

The domain-adversarial neural network (DANN) is a representative deep-learning framework for domain adaptation, originally proposed by Ganin et al. in 2016 [49]. The core idea of DANN is to achieve alignment of source–target feature distributions by embedding adversarial learning within the training process. Specifically, the architecture is composed of three principal modules: a feature extractor, a label classifier, and a domain discriminator. The DANN model is shown in Fig. 1. In the training phase, the feature extractor and label classifier are optimized using the classification loss computed from labeled source-domain samples, which enhances the semantic representation ability of feature extractor network. Concurrently, the domain discriminator is trained with domain labels and a domain-classification loss to separate source and target domains. A gradient reversal layer (GRL) is inserted between the feature extractor and the domain discriminator. This component inverts the gradient direction during backpropagation, thereby driving the feature extractor to learn domain-invariant representations that confound the domain discriminator.

DANN is formulated to minimize the classification loss while maximizing the domain-discrimination loss, yielding a standard min–max adversarial optimization. Through this process, DANN effectively reduces the domain shift and improves the generalization ability on the unlabeled target domain.

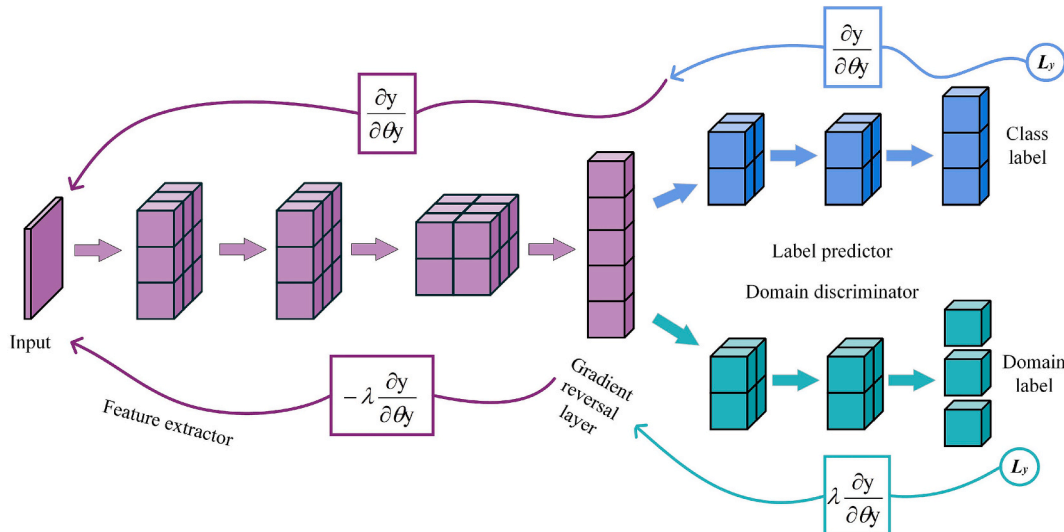


Fig. 1. Classical DANN model.

2.2. Multi-target domain adaptation

In MTDA, knowledge is transferred from a labeled source domain $D_s = \{(x_i, y_i)\}_{i=1}^{n_s}$, drawn from a distribution $P_s(x, y)$, to multiple unlabeled target domains $\mathcal{T}_n = \{D_{t_1}, D_{t_2}, \dots, D_{t_n}\}$, where each target domain $D_{t_k} = \{x_{ik}\}_{i=1}^{n_t}$ originates from a distinct distribution $P_{t_k}(x)$. Although all domains share the same label space, variations in working conditions induce distributional discrepancies, i.e., $P_s(x, y) \neq P_{t_k}(x, y)$, making direct knowledge transfer non-trivial. In MTDA, a domain-invariant representation is learned by projecting the source domain and multiple target domains into a common latent space, thereby reducing inter-domain distribution shift. This enables the model to extend source-domain diagnostic knowledge to all target conditions [50,51]. Effective MTDA strategies must therefore address not only source–target alignment but also mutual alignment among the multiple target domains themselves to reduce inter-domain variance and avoid negative transfer.

2.3. Prototype network

Prototype network was first introduced by Snell et al. at the NIPS conference in 2017 [52], aiming to address few-shot learning problems. As a representative metric-based *meta-learning* framework, the Prototype Network has been extensively employed in tasks such as few-shot classification, cross-domain recognition, and transfer learning. At its core, the method learns an embedding function that maps inputs into a metric space, in which each class is characterized by a prototype vector, typically defined as the centroid of the support-sample embeddings for that class. The architecture of a Prototype Network generally comprises two stages: (1) prototype construction and (2) distance-based classification. During prototype construction, the dataset is partitioned into a support set and a query set. Samples in the support set are grouped by class labels, and each class prototype C_k is computed by averaging the embedded features extracted by a shared encoder f_θ , formulated as:

$$C_k = \frac{1}{|S_k|} \sum_{(x_i, y_i) \in S_k} f_\theta(x_i) \quad (1)$$

where C_k denotes the prototype vector of class k , S_k is the support set of class k , f_θ is the embedding function with learnable parameters θ , and x_i represents a support sample in class k , and y_i denotes its corresponding label.

In the classification phase, each query sample x_q is first mapped into the same embedding space as the prototypes using the shared encoder f_θ . Classification is performed by measuring the distance from the query embedding to each prototype C_k , then applying softmax to the negated distances to obtain a distribution over classes:

$$p(y = k|x_q) = \frac{\exp(-d(f_\theta(x_q), C_k))}{\sum_{j=1}^N \exp(-d(f_\theta(x_q), C_j))} \quad (2)$$

where $d(\bullet, \bullet)$ is a distance metric used to quantify the similarity between the embedded query sample and each prototype, N denotes the number of prototypes, and $d(\cdot, \cdot)$ is chosen as the Euclidean distance in the embedding space.

To optimize the prototype-based classification, the model minimizes the average cross-entropy loss over all query samples. In particular, the probability that a query sample belongs to a specific class is obtained by applying a softmax to the set of negative distances between the sample's embedded feature and the corresponding class prototypes. The training objective is to maximize the likelihood of the ground-truth class, which is achieved by minimizing the following loss function using stochastic gradient descent:

$$\mathcal{L}_p = -\log(p(y = k|x_q)) \quad (3)$$

2.4. Laplace wavelet convolution

Laplace wavelet convolution (LWC) is a trainable convolutional structure inspired by wavelet theory, designed to enhance the sensitivity of neural networks to transient and non-stationary patterns in time-series signals. In tool wear monitoring, force/vibration/AE signals often contain transient bursts and damped oscillations whose scales vary with operating conditions. Compared with standard 1D convolutions, LWC uses Laplace-wavelet-parameterized kernels as a learnable multi-scale band-pass filter bank, yielding a more localized time–frequency representation and improved robustness under condition shifts. Although LWC is not a direct implementation of the continuous wavelet transform (CWT) in the classical sense, its formulation is rooted in the core principles of CWT, particularly its scale–translation invariance and time–frequency localization capability. CWT is a continuous integral transform that decomposes a signal $x(t)$ into a set of basic functions obtained by dilating and translating a mother wavelet $\psi(t)$, as shown in Eqs. (4) and (5).

$$\psi_{a,b}(t) = \frac{1}{\sqrt{a}} \psi\left(\frac{t-b}{a}\right) \quad (4)$$

$$CWT(a, b) = \langle x(t), \psi_{a,b}(t) \rangle = \frac{1}{\sqrt{a}} \int x(t) \psi^*\left(\frac{t-b}{a}\right) dt \quad (5)$$

where $x(t)$ denotes the input signal in the time domain. The parent wavelet $\psi(t)$ serves as the prototype from which all scaled and shifted instances are derived. The basis function $\psi_{a,b}(t)$ is formed by applying a scale $a > 0$ —which governs dilation or compression (small a emphasizes high-frequency detail, whereas large a emphasizes low-frequency content)—together with a translation b that specifies the temporal location. The factor $1/\sqrt{a}$ provides energy normalization across scales, ensuring scale-invariant energy. The notation $\psi^*(\cdot)$ denotes complex conjugation, enabling projections in the complex domain.

However, in practical applications such as vibration-based monitoring or impulse signal analysis, the classical wavelets used in CWT—such as Morlet, Daubechies, and Mexican Hat—are symmetric and oscillatory in both directions, making them less effective in modeling unilateral decaying transients that dominate mechanical impact signals. To address this limitation, the Laplace wavelet is introduced as a one-sided, exponentially decaying complex basis function, better suited for characterizing asymmetric, nonstationary features. The Laplace wavelet is mathematically expressed as:

$$\gamma(t) = A \cdot e^{-\zeta\omega(t-\tau)} \cdot e^{-i\omega(t-\tau)}, t \in [\tau, \tau + W_s] \quad (6)$$

where ω represents the angular frequency of the wavelet. The variable ζ is the damping ratio, which controls how rapidly the wavelet decays in time. The parameter τ determines the time at which the wavelet is centered. The coefficient A is a normalization constant that ensures amplitude consistency. The term W_s defines the time duration over which the wavelet is active. When the damping ratio ζ becomes larger, the wavelet decays more quickly. As a result, the Laplace wavelet becomes more sensitive to high-frequency signal components while naturally suppressing slower, low-frequency variations. Based on this formulation, the LWC is constructed by embedding the Laplace wavelet into a learnable convolutional framework. The convolutional operation is defined as:

$$y(t) = \psi_{a,b,\gamma}(t) * x(t) \quad (7)$$

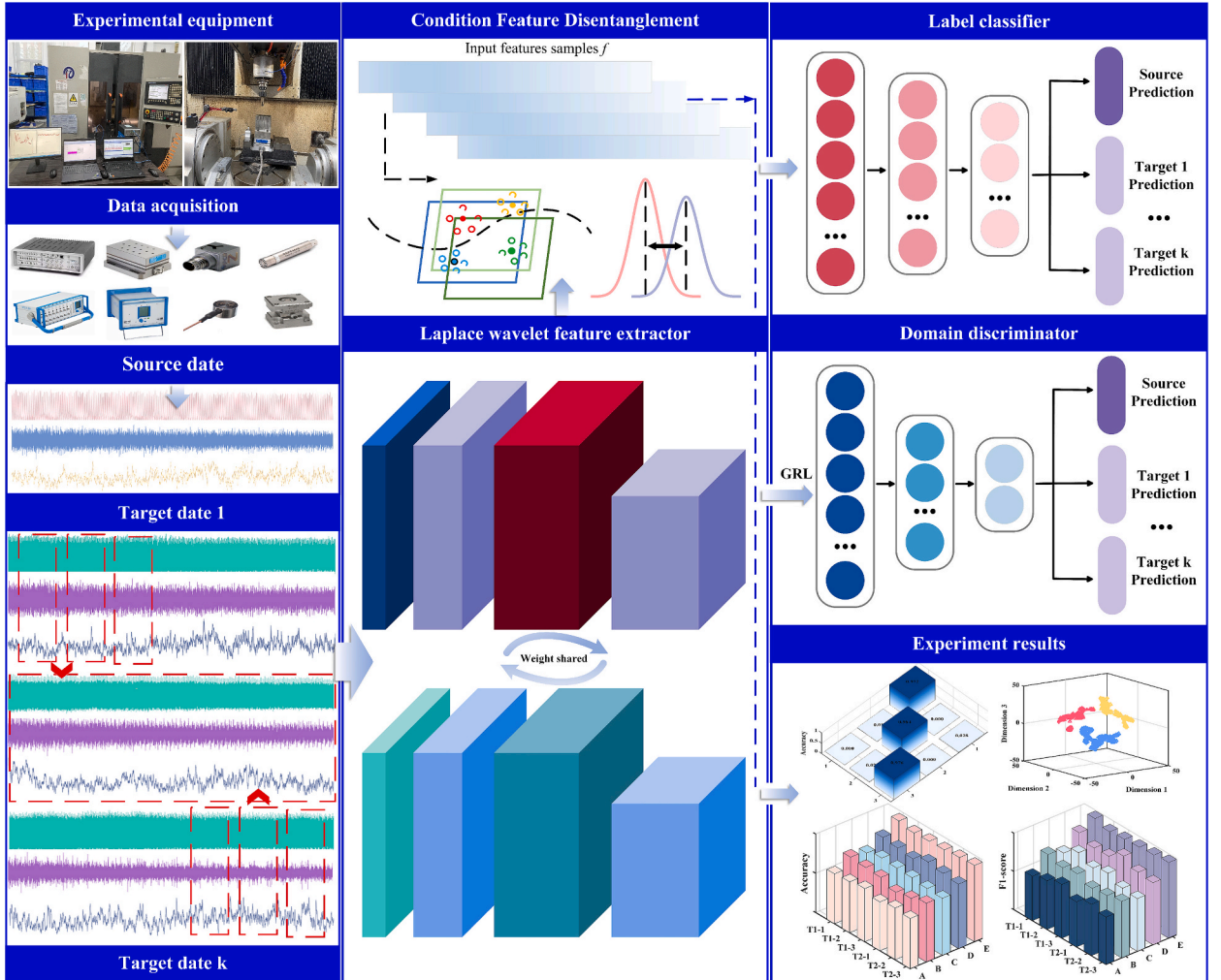


Fig. 2. Framework of the proposed method.

Here, $x(t)$ represents the input signal, while $y(t)$ denotes the corresponding output feature map, and $\psi_{a,b,\gamma}(t)$ is the Laplace wavelet kernel defined by the five learnable parameters: the scale a , translation b , and waveform parameters $\gamma = \{\omega, \zeta, \tau\}$. These parameters are randomly initialized and iteratively updated via backpropagation, enabling the network to adaptively select optimal wavelet characteristics for multiscale feature extraction. Physical interpretability of the Laplace wavelet convolution stems from its functional parameterization. Each kernel is not a free-form weight vector but a Laplace wavelet governed by explicit physical parameters, including the center frequency and the damping factor. Specifically, the center frequency determines the passband location, while the damping factor controls the temporal decay rate and the effective bandwidth, together specifying how the kernel responds to transient excitations. From a machining mechanism perspective, tool wear progression increases frictional excitation and intermittent contact-impact events. These effects are commonly reflected as stage-dependent spectral energy redistribution, including enhanced resonance bands, sidebands, and high-frequency components. During training, optimizing these parameters encourages the kernels to allocate their effective passbands to such informative regions. Therefore, the wavelet front-end can be interpreted as an adaptive filter bank with matched-filter-like behavior, selectively emphasizing wear-sensitive impulse and resonance responses while suppressing slower background variations.

3. The proposed method

To tackle the issue of multi-target domain adaptation under diverse operating conditions, an MCDADAN is proposed in this study. As illustrated in Fig. 2, the model employs a Laplace wavelet convolutional extractor with shared weights across the source domain and multiple target domains to extract wear features, followed by a working condition decoupling module to isolate domain invariant representations. A prototype-guided enhancement classification module is incorporated to improve class separation and ensure reliable supervision, while a multi-discriminator adversarial strategy aligns the source domain with each target domain individually, thereby enhancing generalization performance in cross-domain tool wear recognition tasks.

3.1. Prototype representation and boundary enhancement

To tackle multi-target domain adaptation and strengthen cross-domain classification performance, a prototype-guided enhancement classification and supervision module is proposed. Built upon class prototypes, the module establishes a unified metric space that supports both feature alignment in training and decision support in inference. It integrates three key mechanisms: prototype-based classification, density-aware prototype refinement, and prototype boundary loss for enhancing inter-class separability. These components collectively form a dual-path training structure for unified source-domain supervision and target-domain adaptation.

Let the labeled dataset in the source domain be represented as $D_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$, where x_i^s denotes the i^{th} input sample, and $y_i^s \in \{1, 2, \dots, C\}$ corresponds to its class label among C total categories. Each instance is transformed into a feature vector $f_i^s = f_\theta(x_i^s) \in \mathbb{R}^d$ through a feature extractor $f_\theta(\cdot)$, where d represents the dimensionality of the extracted feature space. The initial prototype C_k for class k is computed as the arithmetic mean of the feature vectors assigned to that class. To mitigate the influence of outliers and edge samples, a density-aware weighting mechanism is introduced. First, compute the squared Euclidean pairwise distances within class k :

$$d_{ij} = \|f_\theta(x_i^s) - f_\theta(x_j^s)\|_2^2, \forall (x_i^s, y_i^s), (x_j^s, y_j^s) \in \mathcal{S}_k \quad (8)$$

where $\mathcal{S}_k$ represents the subset of source samples belonging to class k . Using the average intra-class distance $\bar{d}_k$, define the adjacency matrix $M_k(i, j) = \mathbb{I}(d_{ij} < \bar{d}_k)$. Normalize the adjacency matrix to obtain the sample-wise density weights w_k , and compute the weighted prototype as:

$$C_k = \frac{1}{Z_k} \sum_{(x_i^s, y_i^s) \in \mathcal{S}_k} w_{ki} \cdot f_\theta(x_i^s), Z_k = \sum_i w_{ki} \quad (9)$$

where w_{ki} is the weight assigned to sample x_i^s , and Z_k is a normalization constant. This strategy ensures the class prototype is robust to peripheral noise and better reflects the high-density core representation, thereby enhancing class distinguishability.

Given the prototype C_k , the Euclidean distance from a sample x_i to C_k is defined as:

$$d_{ik} = \|f_\theta(x_i) - C_k\|_2^2 \quad (10)$$

Then, a prototype-driven softmax classifier is constructed:

$$P_i(k) = \frac{\exp(-d_{ik})}{\sum_{j=1}^C \exp(-d_{ij})} \quad (11)$$

For each source sample (x_i^s, y_i^s) , we define a prototype supervision loss that integrates both softmax cross-entropy and intra-class aggregation regularization:

$$L_{pr} = - \sum_{i=1}^{N_i} \log P_i(y_i^s) + \eta \sum_{i=1}^{N_i} \|f_\theta(x_i^s) - C_{y_i^s}\|_2^2 \quad (12)$$

where the first term ensures correct classification, and the second term minimizes intra-class feature dispersion. The trade-off coefficient is set to $\eta = 0.5$.

To sharpen decision boundaries and reduce ambiguity among class prototypes, a prototype boundary loss is introduced. This loss encourages target samples to be closer to their predicted prototype and farther from prototypes of other classes:

$$L_{pb} = \sum_{i \in \mathcal{T}_\tau} \frac{d_{i, \hat{y}_i^t}}{\sum_{j=1}^C d_{ij} - d_{i, \hat{y}_i^t} + \delta}, \delta = 10^{-6} \quad (13)$$

where d_{ij} denotes the distance between the embedding of the target sample x_i^t and the j -th class prototype. Here, T_τ denotes the set of target samples with confident pseudo-labels, selected with $\max_k p(y = k | x_i^t) \geq \tau$, where $\hat{y}_i^t = \arg\max_k p(y = k | x_i^t)$. In this work, τ is set to 0.9 to alleviate noise from low-confidence target predictions. This formulation tends to strengthen prototype-to-sample cohesion for the predicted class while enlarging the separation from competing prototypes.

The final loss function for the prototype-guided classification and supervision module is expressed as:

$$L_P = \lambda_1 L_{pr} + \lambda_2 L_{pb} \quad (14)$$

where λ_1, λ_2 , are weighting coefficients for the respective losses. In this work, we set $\lambda_1 = \lambda_2 = 1.0$.

3.2. Multi-domain alignment

In MTDA, the model needs to simultaneously adapt to multiple target domains with significant distribution differences. In practice, class distributions across targets are often mismatched, with pronounced class overlap and ambiguity. If a uniform alignment strategy is applied—or if target domains are treated indiscriminately—generalization degrades and may even cause classifier collapse. Accordingly, we design a multi-target domain alignment module that enhances the stability and discriminative power of the classifier's outputs while promoting consistency among across domain distributions.

The core challenge in domain adaptation lies in the feature distribution shift between source and target domains, which directly degrades the generalization performance on the target domains. According to the generalization theory proposed by Blitzer et al. [53], the expected target-domain classification error under hypothesis h is upper-bounded as follows:

$$\varepsilon_{\mathcal{T}}(h) \leq \varepsilon_{\mathcal{S}}(h) + \frac{1}{2} d_{H\Delta H}(\mathcal{X}_s, \mathcal{X}_t) + \lambda \quad (15)$$

where $\varepsilon_{\mathcal{S}}(h)$ denotes the empirical error in the source domain, $d_{H\Delta H}$ represents the $H\Delta H$ -divergence that quantifies the discrepancy between the source and target feature distributions under the hypothesis space H , and λ corresponds to the combined error of the optimal joint hypothesis. Nevertheless, these measures fail to account for alignment among the multiple target domains, and thus cannot guarantee simultaneous error minimization across all targets or consistent alignment between each target domain and the source domain. Accordingly, the upper bound of the generalization problem involving multiple target domains can be expressed as in Eq. (16):

$$\sum_{k=1}^K \varepsilon_{\mathcal{T}_k}(h) \leq K \varepsilon_{\mathcal{S}}(h) + \frac{1}{2} \left[\sum_{k=1}^K d_{H\Delta H}(\mathcal{X}_s, \mathcal{X}_{t_k}) + \sum_{(m,n) \in I(K)} d_{H\Delta H}(\mathcal{X}_{t_m}, \mathcal{X}_{t_n}) \right] + K\lambda \quad (16)$$

where m and n represent specific target domain subsets. This extended bound underscores the necessity of aligning source–target distributions and reconciling discrepancies among target domains to achieve global generalization. Additionally, classifier collapse induced by highly correlated inter-class structures further necessitates a robust structural regularization strategy.

3.2.1. Target domain-to-target domain alignment

To minimize the discrepancy among target domains, this work adopts MMD as an explicit alignment criterion. The distributional disparities across these heterogeneous target domains are measured and attenuated, thereby promoting mutual alignment. Given a pair of target domains $(\mathcal{X}_{t_m}, \mathcal{X}_{t_n})$, the MMD is defined as:

$$\begin{aligned} \mathcal{J}_{\text{MMD}}^{(m,n)} &= \left\| \frac{1}{N_m} \sum_{i=1}^{N_m} \phi(x_i^{t_m}) - \frac{1}{N_n} \sum_{j=1}^{N_n} \phi(x_j^{t_n}) \right\|^2 \\ &= \frac{1}{N_m^2} \sum_{i=1}^{N_m} \sum_{j=1}^{N_m} k(x_i^{t_m}, x_j^{t_m}) + \frac{1}{N_n^2} \sum_{i=1}^{N_n} \sum_{j=1}^{N_n} k(x_i^{t_n}, x_j^{t_n}) - \frac{2}{N_m N_n} \sum_{i=1}^{N_m} \sum_{j=1}^{N_n} k(x_i^{t_m}, x_j^{t_n}) \end{aligned} \quad (17)$$

where $\phi(\cdot)$ denotes the mapping into a reproducing kernel hilbert space (RKHS), $k(\cdot, \cdot)$ is the corresponding kernel function, and N_m

and N_n indicate the sample sizes of the respective domains. The overall MMD loss across all domain pairs is given by:

$$\mathcal{L}_{\text{target}} = \sum_{(m,n) \in I(K)} \mathcal{L}_{\text{MMD}}^{(m,n)} \quad (18)$$

By optimizing this loss, the model is driven to learn domain-invariant features common to all target domains, thereby reducing domain bias and alleviating information asymmetry. It is worth noting that target-to-target MMD encourages consistent representations across multiple targets, but if some target domains are highly dissimilar, overly strong mutual alignment may introduce negative transfer by over-smoothing domain-specific structures. In this study, the target domains share the same underlying tool-wear mechanism and differ primarily in cutting parameters, which mitigates this risk. However, for scenarios involving extreme distributional shifts, reducing the MMD weight or adopting selective alignment strategies among subsets of similar domains would be preferable.

3.2.2. Source domain-to-multiple target domain alignment

For each target sample, the current classifier infers its most probable class from the learned feature distribution, enabling alignment between the source domain and multiple unlabeled target domains. Although true labels are unavailable in the target domains, these high-confidence predictions are regarded as pseudo-labels and serve to associate each target sample with its corresponding class prototype derived from the source domain. Through this mechanism, the model performs unsupervised propagation from the source domain to the target domains, as depicted in Fig. 3. For target domain sample x_i^m with same label k , compute its Euclidean distance to the source domain prototype C_k^s to facilitate prototype-level alignment:

$$d_m = \frac{1}{N_k} \sum_{i=1}^{N_k} \|f_\theta(x_i^m) - C_k^s\|_2^2 \quad (19)$$

$$d = \frac{d_1 + d_2 + \dots + d_m}{m} \quad (20)$$

The difficulty of aligning source and target domains is quantified by averaging prototype-space distances between pseudo-labeled target samples and their respective source prototypes. This aggregated distance serves as a measure of domain discrepancy. To further enhance robustness against noise and outliers, a nonlinear weighting function is applied to transform the raw distances. Specifically, the function amplifies the influence of well-aligned samples while down-weighting the contribution of distant or misaligned ones. As a result, the discrepancy score is normalized to the range of (0, 1), where lower values indicate better source-target alignment. This transformed distance serves as a soft alignment weight in the overall domain adaptation objective. To enhance alignment robustness, a nonlinear weighting function is introduced to suppress the influence of distant, potentially noisy sample:

$$\delta = \frac{\exp(\alpha \cdot d) - 1}{\exp(\alpha \cdot d)} \quad (21)$$

This mapping function is monotonically increasing and bounds the output within the range of [0,1). Consequently, a larger distance d yields a δ value closer to 1, representing a significant domain discrepancy, whereas a distance of 0 corresponds to $\delta = 0$, indicating

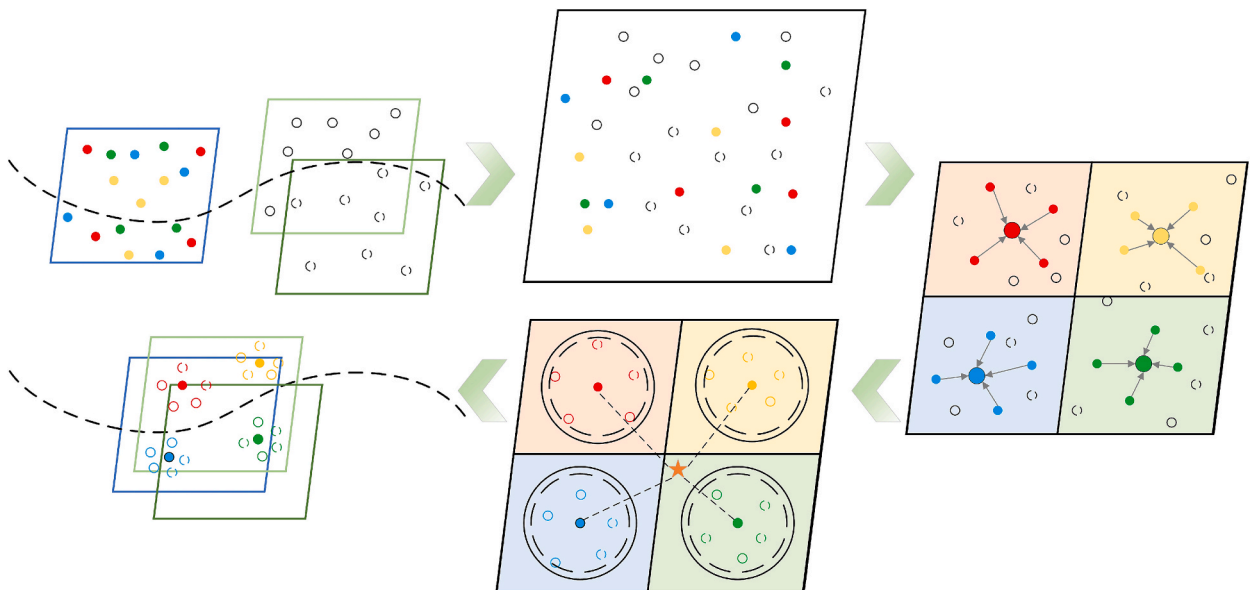


Fig. 3. The source domain is aligned with the multi-target domain.

perfect alignment. The total source-to-target alignment loss is defined as:

$$\mathcal{L}_{\text{source}} = \sum_{m=1}^K \sum_{k=1}^C \delta \quad (22)$$

This mechanism ensures that samples are aligned more closely with their semantic prototypes, reinforcing inter-domain semantic consistency.

3.3. Machining condition variation decoupling

The proposed adversarial transfer learning framework is composed of a shared feature extractor, a label classifier, and a domain discriminator. During the training phase, the label classifier is trained solely on labeled data from the source domain, causing it to become biased toward the source distribution and consequently prone to misclassifying target-domain samples under diverse working conditions. Meanwhile, the domain discriminator is trained using both source and target samples to distinguish domain shifts. This discrepancy in training objectives introduces optimization inconsistency into the feature extractor, potentially causing the extracted representations to entangle domain-relevant information. To alleviate this, a working condition disentanglement module is introduced to explicitly separate domain-relevant and domain-irrelevant components from the shared feature representation. Given an input sample x , the shared feature extractor $f_{\theta}(\bullet)$ produces a latent representation $f = f_{\theta}(x)$, which is then decomposed into domain-relevant and domain-irrelevant components via two independent subnetworks: $f_r = G_r(f)$ and $f_i = G_i(f)$. This disentanglement process is formally expressed as:

$$f_{\theta}(x) \rightarrow (f_r, f_i) \quad (23)$$

To guide the disentanglement, we incorporate learning objectives. A domain classification loss is used to ensure that f_r retains sufficient working condition information. A domain classifier $D_c(\cdot)$ is trained with labeled domain indices. To ensure that f_i is invariant to domain shifts, an adversarial loss is constructed using an additional domain discriminator $D'_c(\cdot)$, trained to distinguish f_i while the feature encoder attempts to fool it:

$$\min_{G_i} \max_{D'_c} \mathbb{E}_x [\log D'_c(f_r) + \log(1 - D'_c(f_i))] \quad (24)$$

A reconstruction decoder $R(\cdot)$ is introduced to enforce semantic consistency, ensuring that the concatenated feature $f_r \oplus f_i$ preserves the original feature semantics by reconstructing the shared feature f . The corresponding reconstruction loss $\mathcal{L}_{\text{rec}}$ in Eq. (25) is used only during training as an auxiliary regularize. During testing, the reconstruction branch is disabled, and the model performs prediction solely using the condition-invariant representation and the classifier, introducing no additional inference overhead:

$$\mathcal{L}_{\text{rec}} = \|R(f_r \oplus f_i) - f\|_2^2 \quad (25)$$

To further suppress condition-sensitive channels and improve invariance, we integrate a channel-wise gating mechanism. The

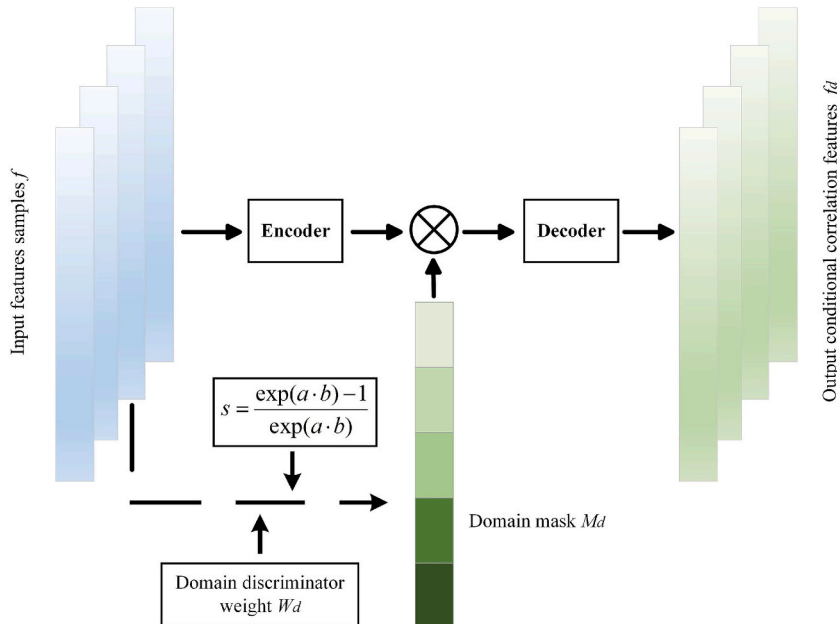


Fig. 4. Machining condition variation decoupling.

global channel response is estimated from the absolute activations of the domain discriminator weight W_d :

$$T_d = \frac{1}{N_h} \sum \sigma(|W_d|) \quad (26)$$

where N_h denotes the number of feature channels, and $\sigma(\cdot)$ represents the activation function. A nonlinear compression factor is computed from the domain discrepancy d to adaptively modulate the threshold:

$$s = 1 - \delta = \exp(-\alpha \cdot d) \quad (27)$$

$$T = T_d \bullet s \quad (28)$$

In this module, the feature representation produced by the shared feature extractor is processed by fully connected encoding and decoding layers, as shown in Fig. 4. The domain mask M_d is then obtained by thresholding:

$$M_d = I(\sigma(|W_d|) > T) \quad (29)$$

Finally, the gated residual transformation is applied to suppress strongly condition-related characteristics:

$$\hat{f}_d = f - g[M_d \cdot W_d \cdot g(W_e f + b_e) + b_d] \quad (30)$$

where W_e , b_e , and b_d denote the parameters of the encoder and the gating path, and $g(\cdot)$ is a nonlinear activation function. It should be explicitly noted that only the resulting condition-invariant representation $\hat{f}_d$ is fed into the subsequent label classifier and multi-domain alignment modules. The condition-relevant component f_r is used only for the auxiliary disentanglement objective during training, so that condition-induced variations are mitigated and wear-discriminative representations are retained. This operation suppresses noisy channels while retaining fault-relevant representations. Feature representations are obtained at the output of the fully connected layers in the encoding stage. These features are subsequently adjusted using a learned gating vector, as described in Eq. (30), which adaptively suppresses condition-sensitive components. This selective filtering enables the extraction of condition-invariant features while preserving essential fault-related information.

3.4. Collaborative optimization objectives

To achieve robust tool wear recognition over multiple target domains, the outputs from the proposed modules are unified within an end-to-end optimization framework. Three core loss functions are formulated based on the condition-invariant feature representations $\hat{f}_d$ obtained from the Machining condition variation decoupling module. Given that labeled data are available in the source domain, health condition classification is directly performed on the condition-invariant features $\hat{f}_d$. A softmax classifier $G_y(\cdot)$ is employed to predict class probabilities, and the cross-entropy loss function is used for supervision:

$$L_y = -\frac{1}{N_s} \sum_{i=1}^{N_s} \log G_y(\hat{f}_d^{(i)})_{y_i^s} \quad (31)$$

where, $\hat{f}_d^{(i)}$ denotes the condition-invariant feature of the i -th source domain sample y_i^s is its ground truth label, and N_s is the total number of source samples. This loss enforces the features to be class-discriminative while enhancing semantic consistency in the source domain.

To encourage extraction of domain-invariant features, we introduce a multi-class domain discriminator $G_d(\cdot)$ that predicts the domain label for each input sample, assigning the source domain label 0 and indexing the target domains from 1 to K . The adversarial loss is defined as:

$$L_d = -\left(\frac{1}{N_s} \sum_{i=1}^{N_s} \log G_d(G_f(x_i^s)) + \sum_{k=1}^K \frac{1}{N_t^k} \sum_{j=1}^{N_t^k} \log(1 - G_d(G_f(x_j^{t_k}))) \right) \quad (32)$$

where x_i^s and $x_j^{(t_k)}$ denote, respectively, samples from the source domain and the k^{th} target domain, and N_t^k indicates the number of samples in target domain k . This objective drives the feature extractor to learn domain-invariant representations that are indistinguishable across multiple domains.

To reduce discrepancies among domains in the feature space, two alignment mechanisms are incorporated: target-to-target domain alignment via MMD, and source-to-target prototype alignment guided by pseudo-labels and enhanced by a nonlinear compression function. The overall discrepancy loss can be formulated as follows:

$$L_b = \alpha \sum_{(m,n) \in \mathcal{J}(K)} \text{MMD}(X^{t_m}, X^{t_n}) + \beta \sum_{k=1}^K \sum_{c=1}^C \delta^{(k,c)} \quad (33)$$

The overall objective function for multi-target domain adaptation, as formulated in Eq. (34), integrates the label classification loss, domain adversarial loss, and domain discrepancy loss. The training procedure is conducted through alternating optimization of the

feature extractor, label classifier, and domain discriminator. The primary goal is to minimize classification errors within the source domain while concurrently reducing distributional discrepancies between the source and target domains, thereby guiding the feature extractor to learn representations resilient to domain shifts.

$$L_{total} = L_y + L_b + \lambda \cdot L_d \quad (34)$$

Although the proposed framework contains multiple objectives, they are designed to operate at different levels and to address different aspects of single-source multi-target tool wear transfer, rather than imposing redundant constraints. Adversarial discrimination mainly performs marginal distribution alignment across domains; however, it does not explicitly preserve class structure and may blur decision boundaries under heterogeneous target shifts. Therefore, the prototype-guided term is introduced as a class-structure-aware regularizer to enhance intra-class compactness and inter-class separability, thereby improving wear-stage discriminability after alignment. In addition, the MMD term is incorporated to encourage inter-target consistency among multiple unlabeled target domains, which is not guaranteed by source–target alignment alone in the multi-target setting. Finally, the machining-condition variation decoupling branch performs representation disentanglement to suppress condition-sensitive components before alignment, while the reconstruction regularizer is used only during training to maintain semantic consistency and avoid trivial solutions. Overall, these objectives are complementary: decoupling reduces condition-driven interference, adversarial learning reduces marginal domain discrepancy, prototype regularization preserves class discriminability, and target-to-target consistency stabilizes multi-target adaptation.

4. Experiments

To evaluate the effectiveness and generalization of the proposed method, two milling wear datasets were used in this study: the PHM2010 milling dataset and a self-constructed GH4169 milling dataset. The PHM 2010 dataset is from the 2010 Prognostics and Health Management (PHM) Challenge and represents typical high-speed milling operations with realistic working conditions, making it a widely recognized benchmark in the field. The GH4169 dataset was collected under various machining conditions, simulating more complex and practical industrial scenarios, particularly in the context of aerospace-grade nickel-based alloy machining. To provide additional evidence of the proposed approach's superiority, it was compared against several widely tool wear monitoring methods.

4.1. Experimental verification for PHM2010 milling dataset

4.1.1. Experimental setup and data preparation

The dataset analyzed in this study is derived from the milling tool wear corpus released by the PHM 2010. As shown in Fig. 5, milling experiments were carried out on a Roders Tech RFM760 high-speed CNC machine using a three-flute carbide end mill to

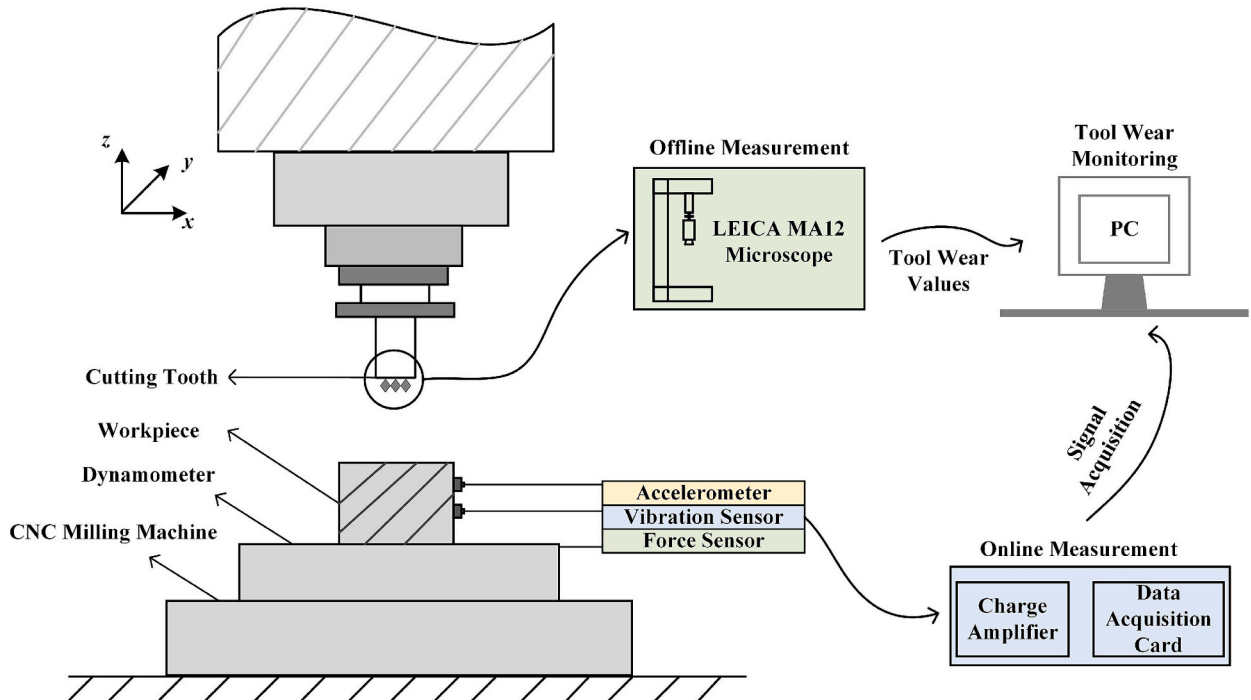


Fig. 5. Schematic diagram of PHM2010 milling experimental platform.

machine Inconel 718 workpieces; detailed machining parameters are summarized in Table 1. To capture tool wear related information, a three-axis dynamometer was mounted on the machine table to record cutting forces along the X, Y, and Z axes. In addition, three piezoelectric accelerometers were affixed to the workpiece to acquire vibration signals in the corresponding directions, and an acoustic emission sensor was employed to capture AE signals during cutting. In total, seven synchronized sensor channels were collected—three for cutting forces, three for vibration, and one for acoustic emission—each sampled at 50 kHz. The analog signals were digitized via a DAQ (NI PCI-1200) for subsequent processing and analysis. During each cutting pass, the tool traversed 108 mm under a fixed machining time, after which the wear of the three cutting edges was measured using a LEICA MZ12 microscope. The seven-channel signal collected during each pass, along with the corresponding wear values, were recorded as an individual sample. The experiment concluded when the tool wear reached a level that rendered the cutter inoperable. For each tool, a total of 315 samples were collected.

The PHM2010 dataset comprises six subsets, designated C1 through C6, among which C1, C4, and C6 contain complete tool life cycle data. Consequently, all subsequent analyses in this study are conducted using these three subsets. As illustrated in Fig. 6, the cutting tool wear curves during the milling process exhibit three distinct phases—initial wear, steady-state wear, and severe wear. To mitigate measurement variations across individual flutes, the wear label for each pass is defined as the average flank wear across the three cutting edges, and this averaged value is used for both training and testing. Notably, substantial variability in wear progression among tools operating under identical cutting conditions leads to class imbalance, particularly in the severe wear stage. To address this issue, Li et al. [18] employed the expectation–maximization algorithm for unsupervised clustering, treating wear measurements as observed variables and wear states as latent variables. This probabilistic clustering approach enables the automatic segmentation of wear states, thereby mitigating class imbalance across the datasets and providing a more uniform data distribution for subsequent analysis. In this study, cutting force, vibration, and acoustic emission signals are used for model training and testing. The classification results of the three balanced subsets are summarized in Table 2. Owing to the high sampling rate of 50 kHz, each sensor channel generates more than 200,000 data points per tool pass, resulting in substantial computational and time costs, as well as potential overfitting. The train/test split is first conducted on the original pass-level samples, and sliding-window segmentation is then performed separately within each subset so that windows from the same pass never appear in both training and testing, preventing information leakage. To improve training efficiency and preserve frequency-domain richness, a non-overlapping sliding window of 10,000 data points is applied to each time-series signal, producing 15 consecutive segments per sample. This non-overlapping strategy controls redundancy under the high sampling rate and avoids generating highly correlated neighboring segments that may bias the multi-target alignment objectives and destabilize pseudo-label propagation. The selected window length is sufficiently long to capture stable wear-related dynamics over multiple tool revolutions, and therefore overlap-based augmentation was not required in this study. This segmentation strategy ensures the continuity and completeness of the tool-cutting signal for subsequent processing and analysis. Furthermore, it transforms raw time-series data into a structure suitable for deep learning models while maintaining temporal dependencies and directional characteristics. The detailed data processing procedure is illustrated in Fig. 7.

4.1.2. Model configuration and implementation details

In this experiment, the computational environment was configured with an Intel Core i7-13700HX CPU and an NVIDIA GeForce RTX 4060 GPU, operating under the Windows 11 system. Model training was performed using PyTorch2.11 with CUDA 12.1 support. To illustrate the fault characteristics under varying working conditions, Fig. 8 presents the time-domain waveforms together with their probability-density distributions. Differences in signal amplitude and distributional form across conditions reveal distinct dynamic responses and tool wear behaviors arising from variations in cutting environments. To ensure a fair comparison and statistical reliability, the hyperparameters of all baseline methods were kept consistent with those of the proposed approach. Each model was evaluated over ten independent runs, and the final results were reported as the average of the remaining outcomes after discarding the highest and lowest values to mitigate stochastic fluctuations. For the proposed method, the network was optimized using stochastic gradient descent (SGD) with a momentum of 0.9, trained for 100 epochs with a batch size of 64 and a learning rate of 0.001. Cutting-force, vibration, and AE signals collected during the milling process served as the primary research data. Each model was executed ten times independently to suppress the effects of stochastic variation. The averaged outcomes were finally reported to provide an accurate assessment of the model's performance and stability.

4.1.3. Comparative methods

To assess performance changes before and after the model improvements, four benchmark models were constructed. To ensure experimental fairness, all models were trained under identical parameter settings:

Model A: The DANN model serves as a representative adversarial domain adaptation framework that leverages a domain discriminator and a gradient reversal layer, exemplifying a classical adversarial transfer learning paradigm.

Table 1
Experimental parameters of the PHM2010 dataset.

Experiment parameters	Values
Spindle speed (n)	10400 rpm
Feed per turn (f)	0.003 mm/r
Cutting depth (a_p)	0.2 mm
Cutting width (a_w)	0.125 mm

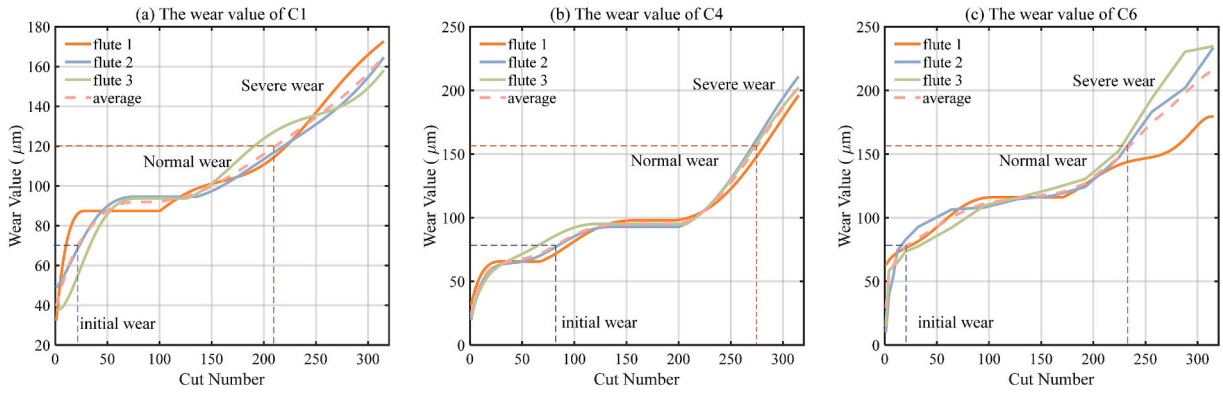


Fig. 6. Cutting tool wear curve.

Table 2

Classification of tool wear states.

Cutting tool	Initial wear	Normal wear	Severe wear
C1	1–47	48–146	147–315
C4	1–135	136–204	205–315
C6	1–81	82–188	189–315

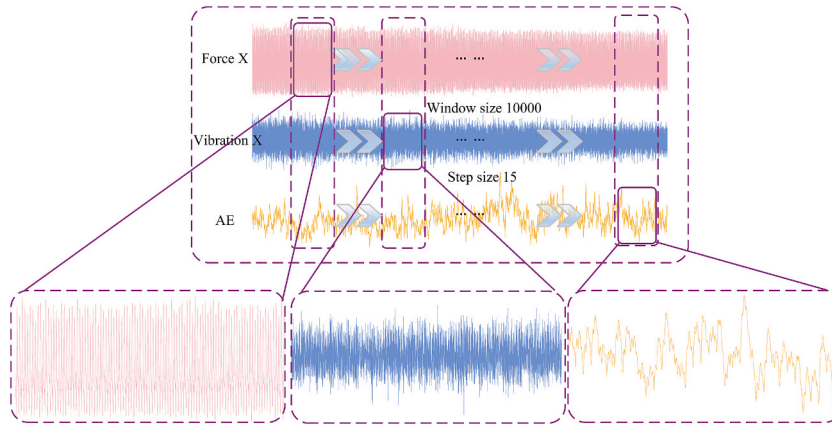


Fig. 7. Data preprocessing schematic diagram.

Model B: The DSAN model integrates a dual semantic alignment mechanism that simultaneously extracts common semantic representations and aligns subdomain-level features between the source and multiple target domains.

Model C: The machining-condition variation decoupling module is employed to extract features independent of specific working conditions; however, it lacks a prototype-enhancement component to guide class-level feature alignment across domains.

Model D: A prototype enhancement module is utilized to align class-level features between source and target domains, but there is a lack of a Machining condition variation decoupling module that extracts domain-invariant representations under varying working conditions.

Model E: The method proposed in this study.

4.1.4. Parameter determination

To measure the distributional discrepancy between the source domain and each target domain, this study adopts the nonlinear mapping function defined in Eq. (27). As shown in Fig. 9(a), smaller values of α produce a more gradual increase, allowing minor domain discrepancies to be preserved, while larger values result in a sharper rise, emphasizing significant inter-domain differences. By tuning α , the model adaptively redistributes weights based on domain distances, thereby suppressing irrelevant domain-specific variations and enhancing the learning of invariant fault features.

In this study, α is set to [0.001, 0.005, 0.01, 0.02, 0.05, 0.08, 0.1, 0.5] for domain-level experiments of scaling on feature mapping. A domain adaptation experiment was performed from the source domain C1 to the target domains C4 and C6 using the Machining

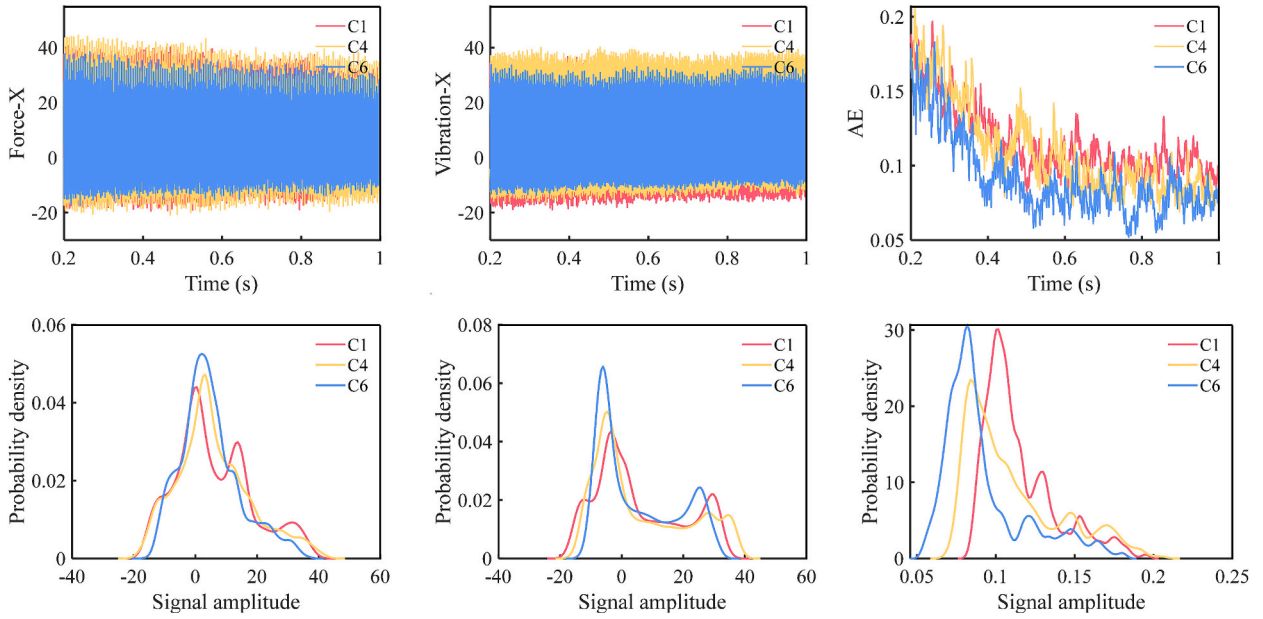


Fig. 8. Three signals and probability density distribution of signals.

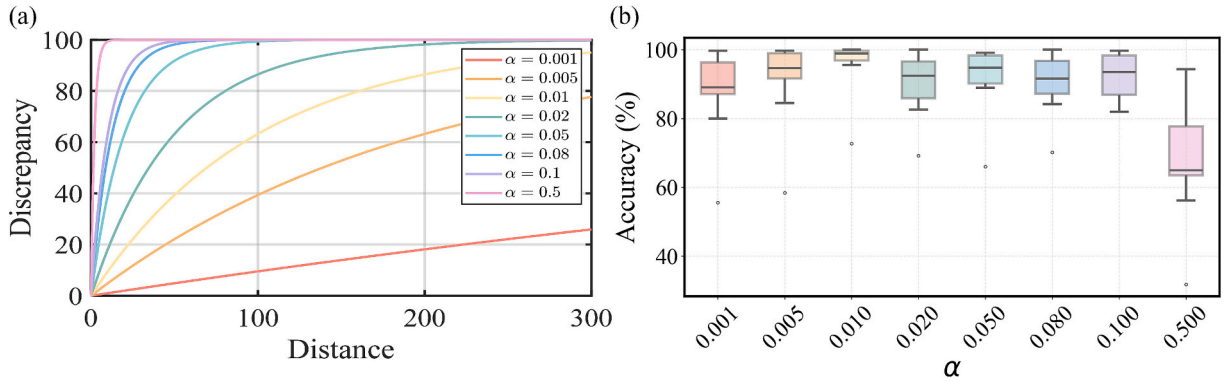


Fig. 9. Influence of the threshold parameter α on discrepancy mapping and model accuracy.

condition variation decoupling module alone, with the training process lasting for 100 epochs. The average diagnostic accuracy across the two target domains was subsequently calculated and plotted in Fig. 9(b). As illustrated in Fig. 9(b), the parameter α has a critical effect on diagnostic accuracy in the tool wear monitoring. When α increases from 0.001 to 0.01, the accuracy rapidly improves, reaching the highest median at $\alpha = 0.01$ with minimal variance. For α between 0.05 and 0.1, the accuracy remains relatively high but exhibits growing instability and more outliers. Once α reaches 0.5, the model's performance drops sharply, with the median falling to around 65% and a wide range extending down to 30%, indicating over-regularization that disrupts the model's discriminative ability. In summary, $\alpha = 0.01$ achieves the best trade-off between diagnostic performance and stability and this study identifies $\alpha = 0.01$ as the optimal threshold parameter.

The balance between aligning the source domain with each target domain and maintaining mutual consistency among target domains is crucial for achieving optimal performance in multi-target domain adaptation. Overemphasizing MMD may cause excessive alignment among target domains, leading them to deviate from the source distribution and diminishing model generalization. Conversely, enforcing overly strong source–target alignment through a prototype-based strategy can introduce inconsistencies among target domains, thereby hindering across targets. Hence, appropriate tuning of the corresponding penalty coefficients is essential. As depicted in Fig. 10, when both α and β are set to zero, the average accuracy across target domains drops to 87.14%, demonstrating that the feature-alignment modules are indispensable to overall model performance. In particular, when $\beta \geq 0.6$, the average accuracy consistently exceeds 90% regardless of the value of α , suggesting that the two coefficients are not simply linearly dependent and must be tuned jointly. The highest accuracy is achieved at $\alpha = 0.5$ and $\beta = 0.6$, which is thus selected as the optimal configuration. This pattern reflects a nonlinear relationship between the two penalties. Increasing either one can enhance performance up to a saturation point, beyond which over-alignment degrades discriminative capability. Taken together, these results indicate that achieving robust,

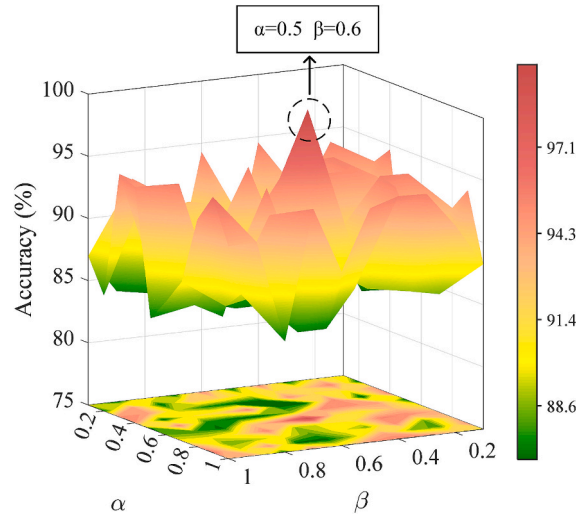


Fig. 10. Sensitivity of proposed method to tradeoff coefficients α and β .

generalizable domain adaptation requires a calibrated trade-off between the strength of source-target alignment and the consistency maintained across target domains.

As shown in Fig. 11, the mean accuracy across target domains exhibits a slight decline over the first nine epochs, a behavior attributable to early-stage mutual alignment among target domains enforced by the MMD constraint. This phase reflects the model's attempt to reduce inter-target domain divergence before engaging in cross-domain alignment. From epoch 10 to 18, a rapid rise in accuracy is observed, driven by the joint effect of MMD and prototype-guided learning, which effectively reduces domain-level discrepancies and enhances category-level feature alignment. Beyond epoch 18, the source-target similarity continues to improve and gradually saturates, while the target-target discrepancy decreases sharply and stabilizes, indicating that both inter-domain consistency and intra-class compactness are progressively reinforced. It is noteworthy that the model attains its maximum accuracy near epoch 89, coinciding with the epoch at which the similarity between the source domain and each target domain reaches its peak. Meanwhile, the total loss shows a sharp decline in the early training stage and converges to a stable, low level, reflecting smooth optimization and convergence. The concurrent trends of increasing accuracy, enhanced domain similarity, and reduced discrepancy collectively validate the proposed method's capacity to alleviate alignment bottlenecks and enable consistent cross-domain generalization.

4.1.5. Experimental verification and analysis

In the PHM2010 dataset, vibration, force, and AE signals are collected during the tool wear process under different machining conditions. To comprehensively evaluate the effectiveness of the proposed method, three cross-domain transfer experiments were performed under varying working conditions, as summarized in Table 3. Specifically, three single-source domain and multi-target domain transfer tasks were constructed by combining distinct tool wear states and machining environments. In this paper, a domain is defined as a dataset collected under a consistent machining scenario. For the PHM2010 dataset, although C1, C4, and C6 correspond to different cutters with essentially identical nominal cutting parameters, the inherent tool-to-tool variability and run-to-run factors lead to distinct feature distributions. Therefore, each cutter is technically treated as a separate domain. To ensure a rigorous evaluation, we adopted a round-robin strategy where each tool serves as the source domain in turn. This design validates that the

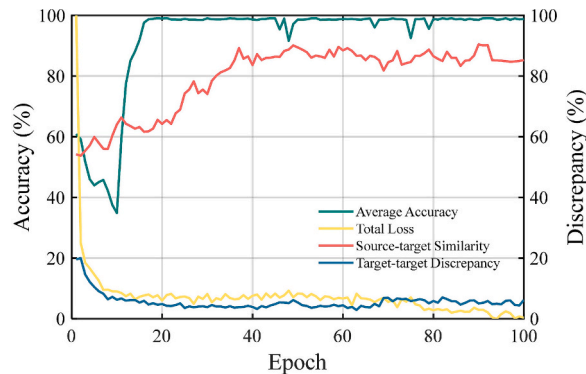


Fig. 11. Training of the proposed model.

Table 3
PHM2010 dataset wear experiment.

Experimental task	Source domain	Target domain
Task 1	C1	C4, C6
Task 2	C4	C1, C6
Task 3	C6	C1, C4

model's generalization capability is robust to stochastic physical variability and is not dependent on the specific characteristics of a single source tool. For each task, the trained model produced the average diagnostic accuracy across all target domains, along with precision, recall, and F1-score metrics for each target domain. These indicators collectively assess the diagnostic capability of the model.

From the diagnostic accuracy results of the PHM2010 wear experiments shown in Fig. 12, several conclusions can be drawn. The proposed model achieves an average diagnostic accuracy of 97.63%, exhibiting the smallest accuracy fluctuation among the three cross-domain transfer tasks. This outcome demonstrates that Model E possesses strong robustness to domain shifts, maintaining superior and stable diagnostic performance compared with other approaches. By examining Model A against other domain adaptation methods, Model A performs best in Task 1—where the source domain is C1 and the target domains are C4 and C6—since the source domain lies between the two target domains in feature distribution space. However, in other transfer scenarios where significant discrepancies exist between the target domains, Model A tends to bias toward one domain, suggesting that while adversarial domain adaptation enhances diagnostic performance under complex feature distributions, it can also introduce domain-specific bias when inter-domain divergence is large. A comparison between Model C and Model D reveals that Model D, which incorporates an improved prototype network, surpasses Model C in Task 3, indicating that aligning each target domain with the source domain via an enhanced prototype network can improve diagnostic accuracy. Nonetheless, in transfer tasks characterized by substantial variations in operating conditions, Model C yields higher diagnostic accuracy than Model D. This finding suggests that the machining-condition variation decoupling module effectively filters condition-dependent interference, thereby improving cross-domain diagnostic generalization.

To evaluate the effectiveness of feature extraction and domain alignment in Task 1, this work apply t-distributed stochastic neighbor embedding (t-SNE) is applied to project the high-dimensional features into a lower-dimensional space, as depicted in Fig. 13. Model A and Model B exhibit ambiguous boundaries between wear categories, particularly between Initial Wear and Normal Wear. The scattering of same-class points across domains suggests weak cross-domain alignment and limited classification robustness. In contrast, Model C achieves better intra-class compactness and clearer separation among different wear levels. However, inconsistencies remain in the feature alignment of identical classes between the two target domains, indicating room for improvement in domain-invariant representation learning. Model D further enhances category discrimination, with noticeable improvements in both class separability and domain consistency. In contrast, the proposed method enables the features from the source domain to overlap significantly with those from both target domains, while preserving clear boundaries between different wear categories. This observation suggests that the proposed approach effectively mitigates domain discrepancies, enabling diagnostic knowledge learned from the source domain to generalize comprehensively across all target domains, thereby exhibiting strong diagnostic capability and cross-domain generalization performance.

To further assess classification performance under the multi-target domain adaptation setting, Fig. 14 reports the confusion matrices for Target Domain 1 and 2. As illustrated in Fig. 14(a) and (b), Model A exhibits significant confusion across the three wear types, particularly with Initial wear samples being frequently misclassified as Normal or Severe, indicating limited category discriminability. Although Model B (Fig. 14(c)–(d)) shows slightly improved accuracy, it still struggles to distinguish Initial and Severe

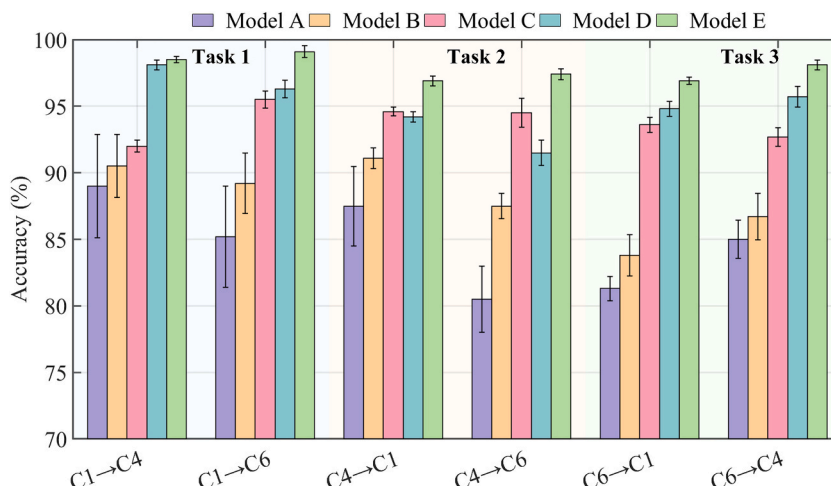


Fig. 12. Diagnostic result in PHM2010 dataset wear experiment.

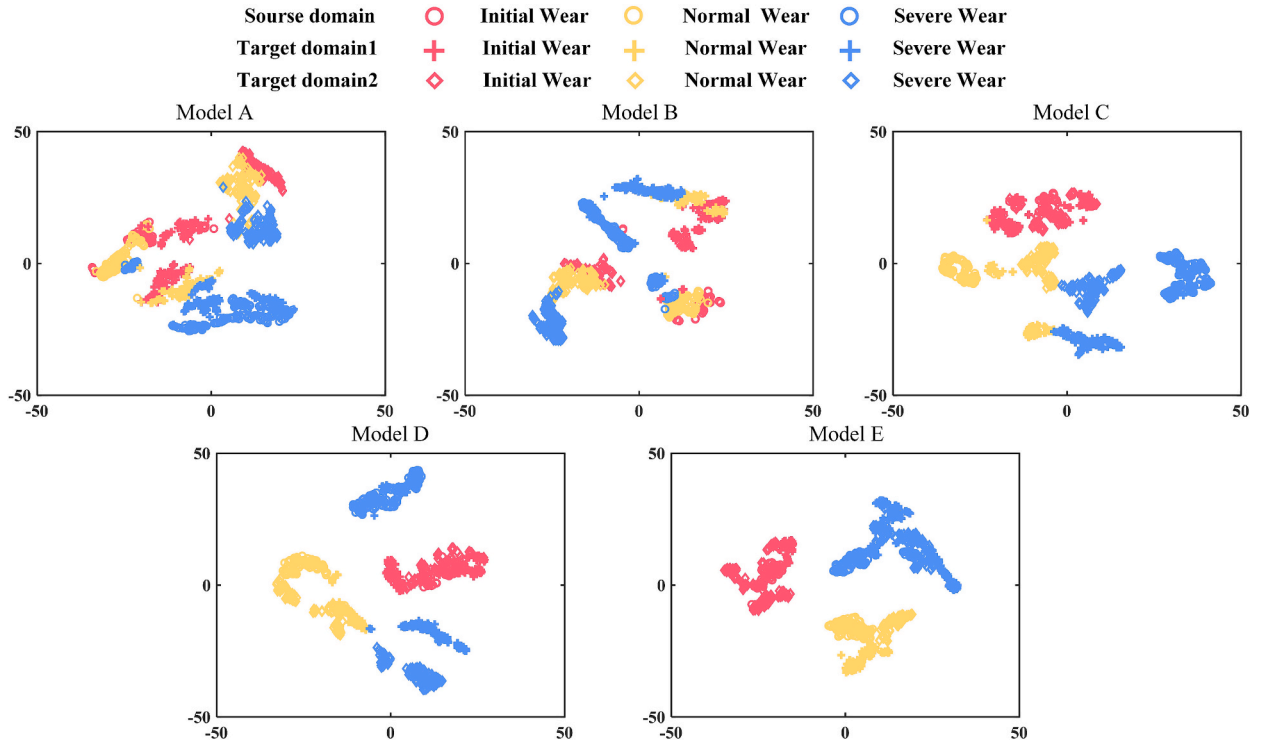


Fig. 13. Visualization of t-SNE features for the Task 1.

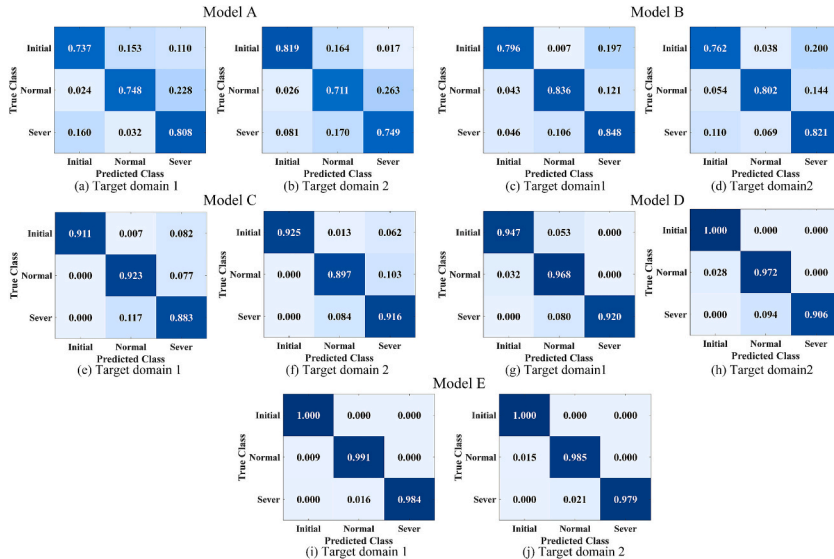


Fig. 14. Confusion matrices of multi-target domain samples in Task 1.

wear states effectively. Model C and Model D, incorporating prototype alignment and working condition disentanglement mechanisms, respectively, achieve more stable recognition across both domains. Notably, Model D demonstrates robust classification without heavy reliance on source domain guidance, accurately identifying all wear states even under varying conditions. A qualitative inspection of Task 1 suggests that many errors are absorbed into the Normal state: Initial wear may share tooth-passing-dominated spectral characteristics with Normal wear and thus exhibit overlapping amplitudes, whereas Severe wear can manifest intermittently as impulsive bursts and broadband high-frequency energy that is not consistently present in every segment, making it easier to be confused with Normal wear. Finally, Model E, representing the proposed method, achieves nearly perfect performance in both target domains, with recognition rates exceeding 97% for all wear categories. This result confirms its superior capability in capturing

discriminative features and enhancing generalization under complex working conditions. In addition, precision, recall, and F1-score are used to assess model performance. As summarized in Table 4, suboptimal performance in recognizing the severe wear condition is observed in Model A, with an F1-score of only 0.738 in Target Domain 2, suggesting a risk of misclassifying critical wear states as normal, which poses potential challenges to practical manufacturing reliability. Improved classification performance across all wear states is observed in Model B after the incorporation of a dual semantic alignment module, especially regarding recall and F1-score. For Model C, where a Machining condition variation decoupling mechanism is introduced, recognition performance for severe wear is further enhanced, with F1-scores of 0.865 and 0.880 being yielded in the two domains, respectively. In Model D, where a prototype enhancement module is utilized, strong performance is demonstrated, especially for initial and severe wear, with F1-scores of 0.957/0.921 and 0.958/0.951 being recorded in Target Domain 1 and 2. The best overall performance is achieved by the proposed Model E, with all F1-scores exceeding 0.985, and scores of 0.992 and 0.989 being reached for severe wear classification. These results indicate that the model demonstrates strong performance and computational efficiency in practical applications.

4.2. Experimental verification for milling of GH4169

4.2.1. Machining setup and conditions

With the increasing demand for aerospace engines and gas turbines, high-temperature alloys have seen wider application despite their inherently poor machinability, underscoring the importance of reliable tool wear monitoring. To further verify the accuracy and robustness of the proposed method in tool wear monitoring, milling experiments were carried out using GH4169, a representative nickel-based superalloy. A series of dry milling tests were conducted under varying cutting parameters on a five-axis CNC machining center. The experimental configuration and instrumentation setup are depicted in Fig. 15. The cutting tool employed was a two-flute indexable end mill, model EMP01-020-G20-AP11-02, with a diameter of 20 mm, fitted with APKT11T304-APL inserts. The workpiece material, GH4169, measured 100 mm × 100 mm × 100 mm, with each cutting path spanning 100 mm. The physical properties of the workpiece and the corresponding tool parameters are summarized in Table 5.

4.2.2. Experimental verification and analysis

To achieve comprehensive monitoring and overcome the limitations of single-sensor measurements, four types of sensors were integrated to capture multi-source data throughout the cutting process. These included cutting force, vibration acceleration, acoustic emission, and sound signals. Cutting forces were measured using the Kistler 92 measurement system, comprising a multi-component dynamometer (model 9257B), a charge amplifier (model 5070A), and a data acquisition card (model 5697A). Vibration signals generated during machining were recorded via a PCB356A15 accelerometer mounted on the spindle, while sound signals were collected using a GRAS 46AE microphone positioned on the machine base. Both vibration and sound data were acquired through the LMS data acquisition system. Additionally, an acoustic emission sensor (model PAC R15α) was installed on the workpiece sidewall to capture AE signals. The cutting parameters and cross-experimental configurations are summarized in Tables 6 and 7, respectively. To thoroughly evaluate the proposed method's performance, seven single-source cross-domain transfer experiments were conducted. For the GH4169 dataset, each domain explicitly corresponds to a fixed machining-condition tuple (n, f, a_p, a_e) under an invariant setup, meaning that any variation in these physical parameters forms a new domain. In the first six tasks, two working conditions served as target domains, while the final task included three target-domain conditions. These source–target combinations were designed to reflect practical production scenarios involving process adjustments. Accordingly, the tasks cover a range of distribution shifts, from primarily single-parameter changes to combined multi-parameter changes, thereby testing the model's adaptability across the machining-parameter space. For each task, the trained model produced diagnostic accuracy metrics for all target domains, which were used as the key indicators to assess diagnostic performance.

Table 4
Model experiment results in Task 1.

Numerical (%)		Assessment indicators					
		Target domain 1			Target domain 2		
		Precision	Recall	F1-score	Precision	Recall	F1-score
Model A	Initial wear	0.800	0.737	0.767	0.884	0.819	0.850
	Normal wear	0.802	0.748	0.774	0.680	0.711	0.695
	Severe wear	0.705	0.808	0.753	0.728	0.749	0.738
Model B	Initial wear	0.899	0.796	0.845	0.823	0.762	0.791
	Normal wear	0.881	0.836	0.858	0.882	0.802	0.840
	Severe wear	0.727	0.848	0.783	0.705	0.821	0.758
Model C	Initial wear	1.000	0.911	0.953	1.000	0.925	0.961
	Normal wear	0.882	0.923	0.902	0.902	0.897	0.900
	Severe wear	0.847	0.883	0.865	0.847	0.916	0.880
Model D	Initial wear	0.967	0.947	0.957	0.973	1.000	0.986
	Normal wear	0.879	0.968	0.921	0.912	0.972	0.941
	Severe wear	1.000	0.920	0.958	1.000	0.906	0.951
Model E	Initial wear	0.991	1.000	0.996	0.985	1.000	0.993
	Normal wear	0.984	0.991	0.988	0.979	0.985	0.982
	Severe wear	1.000	0.984	0.992	1.000	0.979	0.989

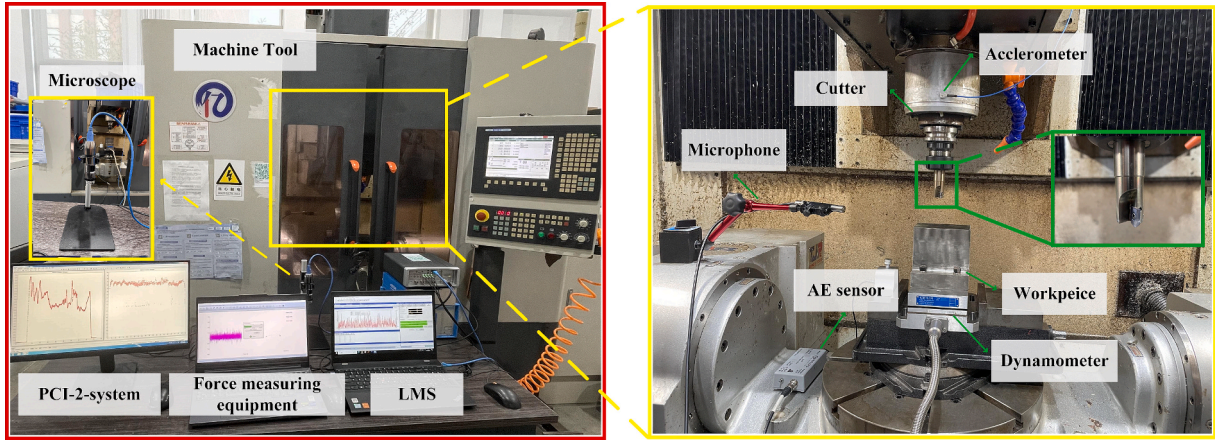


Fig. 15. Experimental setup of tool wear monitoring.

Table 5
Parameters of the workpiece and tool.

Tool	Parameters	Workpiece	Parameters
Coating	PVD-TIALN	Material	GH4169
Insert thickness (mm)	3.6	Density (kg/m^3)	8280
Cutting edge length (mm)	12	Yield strength (MPa)	1275
Corner radius (mm)	0.4	Tensile strength (MPa)	1430
Clearance angle major ($^\circ$)	11	Elongation (%)	20

Each cutting experiment began with the installation of a new insert. During machining, the insert was periodically removed from the tool holder and its flank wear was measured using an AO-3 M180 microscope until tool failure occurred. According to the manufacturer's standard, 0 ~ 0.1 (mm) is initial wear, 0.1 ~ 0.3 (mm) is normal wear, and greater than 0.3 (mm) is severe wear. Fig. 16 illustrates three different tool wear states under the milling condition Case 1. During the machining of GH4169 and other typical high-temperature alloys, limited thermal conductivity restricts the dissipation of cutting heat, causing thermal accumulation at the tool-workpiece interface. This thermal concentration promotes the formation and stable adhesion of built-up edges on the workpiece surface. The repetitive interaction between these built-up edges and the cutting tool markedly accelerates tool wear, especially along the flank face near the axial depth of cut. Consequently, in this study, flank wear at the axial cutting depth is adopted as the labeling criterion to ensure accurate identification of tool wear states. To further validate the superiority of the proposed method, experimental verification based on the GH4169 milling dataset demonstrates that the MCDADAN model achieves clear advantages across key evaluation metrics. The diagnostic accuracies for different transfer tasks are summarized in Table 8, and the comparative results for each task are presented in Fig. 17. As observed in Table 8, Model E consistently outperforms the other approaches across six cross-

Table 6
Milling experimental parameters.

Condition case	Spindle speed (rpm)	Feed speed (mm/min)	Cutting width (mm)	Cutting depth (mm)
Case 1	1000	100	0.1	1.5
Case 2	1000	200	0.2	1.5
Case 3	1200	240	0.2	1.5
Case 4	800	240	0.3	1.5

Table 7
GH4169 dataset wear experiment.

Experimental test	Source domain	Target domain
Test 1	Case 1	Case 2, Case 3
Test 2	Case 2	Case 1, Case 4
Test 3	Case 3	Case 1, Case 2
Test 4	Case 4	Case 2, Case 3
Test 5	Case 2	Case 3, Case 4
Test 6	Case 3	Case 1, Case 4
Test 7	Case 1	Case 2, Case 3, Case 4

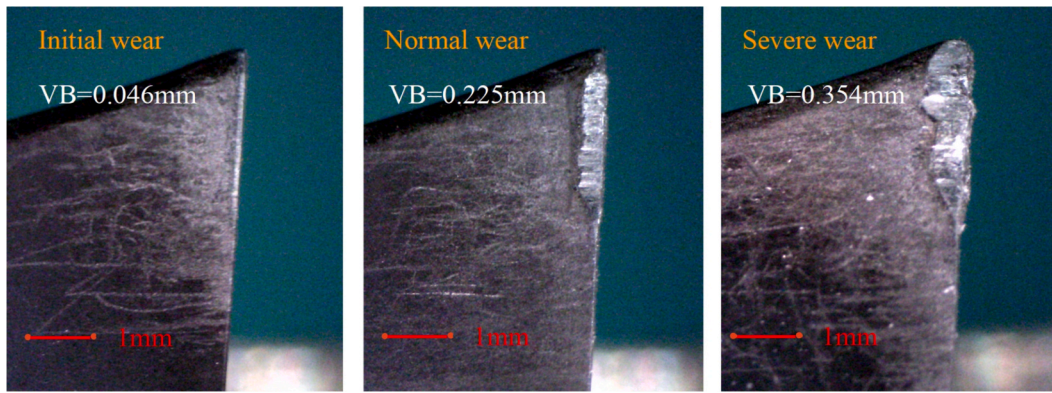


Fig. 16. Different tool wear states under cutting case1.

condition transfer tasks, attaining diagnostic accuracies exceeding 97% in nearly all cases. Specifically, it achieves 97.31% in Test 1 and 98.14% in Test 2, with standard deviations remaining around 0.6%, indicating excellent stability and adaptability. These results highlight the model's capability to extract domain-invariant yet highly discriminative features under diverse operating conditions, making it particularly suitable for complex machining environments characterized by frequent variations in load and spindle speed. Model D also delivered strong diagnostic performance, reaching 95.75% in Test 2 and 96.96% in Test 5. It surpassed Model C in most tasks and clearly outperformed traditional adversarial domain adaptation approaches. Although its accuracy approached that of Model E, it exhibited slightly greater performance variability, indicating lower robustness under certain condition shifts. Model C achieved mid-to-high overall accuracy, with results of 95.58% in Test 3 and 94.42% in Test 6. However, it showed sensitivity to larger domain discrepancies; for instance, accuracy dropped to 91.36% in Test 1, and standard deviations exceeded 2% in several tasks, reflecting limited stability. In contrast, Models A and B showed clearly inferior performance. Model A, in particular, reached only 70.95% accuracy in Test 5, which was the lowest accuracy recorded among all models. Moreover, it exhibited considerable fluctuations, indicating significant limitations in handling nonstationary transfer tasks and reduced suitability for tool wear monitoring under complex machining conditions.

As illustrated in Fig. 17, the proposed method consistently outperforms all comparative approaches across the six transfer tasks in terms of classification accuracy. This performance advantage evidences the method's efficacy in markedly mitigating the domain gap, thereby supporting reliable transfer and broad generalization of diagnostic knowledge learned in the source domain across diverse target domains. The method exhibits strong discriminative power for fault features and excellent generalization capability. Furthermore, as shown in Fig. 18, the radar chart provides an intuitive visualization of the standard deviation across different transfer tasks. The MCDADAN method achieves the lowest and most stable standard deviation among all methods, highlighting its exceptional robustness under domain shifts. To further comprehensively evaluate the performance of MCDADAN, an extended experimental setting was conducted beyond the six single-source cross-domain transfer tasks. Specifically, a new scenario involving one source domain and three target domains was introduced. As summarized in Table 8, the average diagnostic accuracy of all models decreased with the increased complexity introduced by three target domains. Model A showed a notable decline in performance, achieving an average accuracy of 71.41%, representing a reduction of 21.91%. In contrast, the proposed MCDADAN method maintained high performance, reaching an average accuracy of 93.32%. It surpassed Models A through D by 21.91%, 13.91%, 5.87%, and 3.39%, respectively. To more intuitively demonstrate the superiority of the proposed model in domain adaptation, the learned domain-invariant features were visualized using t-SNE, a nonlinear dimensionality reduction technique that projects high-dimensional representations into a two-dimensional space. As illustrated in Fig. 19, the clustering results of five models are displayed for transfer task Test 7, where the source domain corresponds to Case 1 and the target domains correspond to Case 2, Case 3, and Case 4. As shown in Fig. 20, the fault features extracted by Model A exhibit severe overlap across categories, leading to frequent misclassifications among early wear, stable wear, and severe wear states. This indicates poor discriminative capability of the model. Although Model B achieves a certain degree of separation among the three wear types, its learned deep feature mappings show significant overlap between early wear and stable wear, lacking clear linear separability. This largely explains its low diagnostic accuracy under varying operating conditions. Compared with the previous models, both Model C and Model D significantly enhance the wear classification capability by increasing inter-class distances and reducing intra-class dispersion, thereby improving the clustering effect. However, under different operating conditions, samples of the same wear category still exhibit poor alignment across target domains, indicating that the feature representations lack sufficient consistency and limiting the model's generalization performance in multi-condition scenarios. In contrast, the proposed MCDADAN model yields feature with superior linear separability and higher inter-class distinguishability, characterized by compact intra-class clusters and well separated class boundaries. This facilitates accurate and resilient cross-domain wear monitoring over a broad spectrum of operating conditions.

To further substantiate the method's effectiveness for cross-domain monitoring under diverse operating conditions, the diagnostic results of each model in Test 7 are visualized using confusion matrices, as shown in Fig. 20. The five models exhibit markedly different performance across the three target domains. Model A and Model B achieve average classification accuracies of 71.97% and 75.83%, respectively, both showing substantial misclassification, particularly in identifying severe fault categories under varying operating

Table 8
Model accuracy in GH4169 dataset wear experiment.

Source domain	Case 1 (Test 1)		Case 2 (Test 2)		Case 3 (Test 3)	
Target domain	Case 2	Case 3	Case 1	Case 4	Case 1	Case 2
Model A	84.62 ± 4.69	84.21 ± 5.16	81.40 ± 4.67	81.08 ± 4.53	76.84 ± 4.31	78.41 ± 4.36
Model B	83.43 ± 4.13	83.36 ± 3.91	82.57 ± 3.48	83.26 ± 3.35	80.10 ± 3.12	79.62 ± 2.98
Model C	91.36 ± 2.05	93.58 ± 1.99	92.73 ± 2.31	91.60 ± 2.21	93.01 ± 1.17	95.22 ± 1.18
Model D	90.39 ± 0.79	90.16 ± 0.91	94.51 ± 0.93	95.75 ± 0.45	96.51 ± 0.46	97.01 ± 0.69
Model E	97.31 ± 0.58	97.44 ± 0.52	98.14 ± 0.46	98.41 ± 0.49	97.37 ± 0.18	98.37 ± 0.47
Source domain	Case 2 (Test 4)		Case 2 (Test 5)		Case 3 (Test 6)	
Target domain	Case 3	Case 4	Case 3	Case 4	Case 1	Case 4
Model A	75.52 ± 5.00	75.82 ± 4.78	81.87 ± 3.97	70.95 ± 4.16	81.91 ± 4.16	81.79 ± 4.17
Model B	83.13 ± 2.98	82.64 ± 3.03	86.62 ± 3.34	88.10 ± 3.16	84.36 ± 3.05	85.57 ± 3.15
Model C	93.84 ± 2.08	92.75 ± 2.12	95.58 ± 1.09	95.94 ± 1.54	94.42 ± 1.49	94.67 ± 1.53
Model D	97.79 ± 1.03	97.56 ± 1.28	94.67 ± 1.76	96.96 ± 1.72	93.28 ± 0.87	94.71 ± 0.64
Model E	98.31 ± 0.42	97.77 ± 0.23	96.33 ± 1.21	97.44 ± 1.19	98.11 ± 0.52	98.21 ± 0.48
Source domain			Case 1 (Test 7)			
Target domain			Case 2	Case 3	Case 4	
Model A			70.52 ± 5.06	71.95 ± 5.11	71.77 ± 4.80	
Model B			80.01 ± 4.47	79.62 ± 4.29	78.61 ± 4.24	
Model C			87.93 ± 3.29	87.64 ± 3.25	86.79 ± 3.17	
Model D			89.60 ± 3.06	90.22 ± 2.83	89.96 ± 2.94	
Model E			94.42 ± 1.41	92.16 ± 1.53	93.39 ± 1.60	

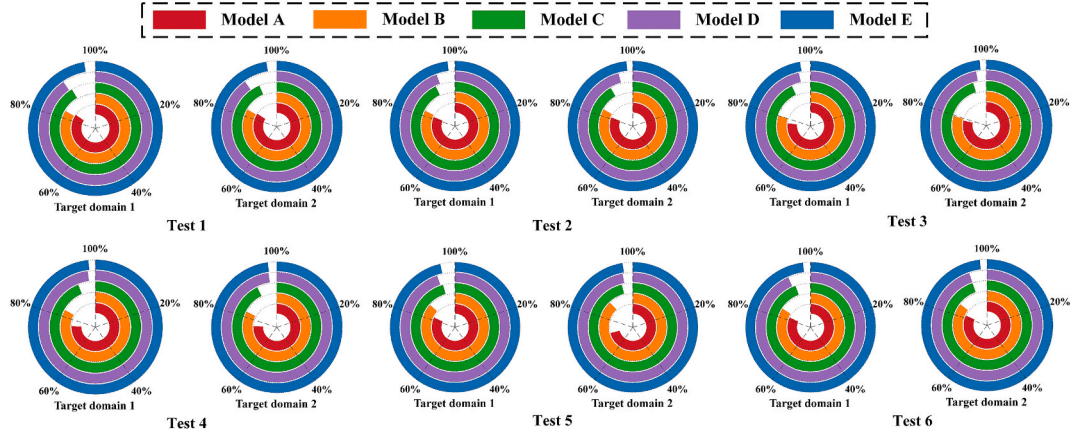


Fig. 17. Comparison of diagnostic accuracies across six test groups.

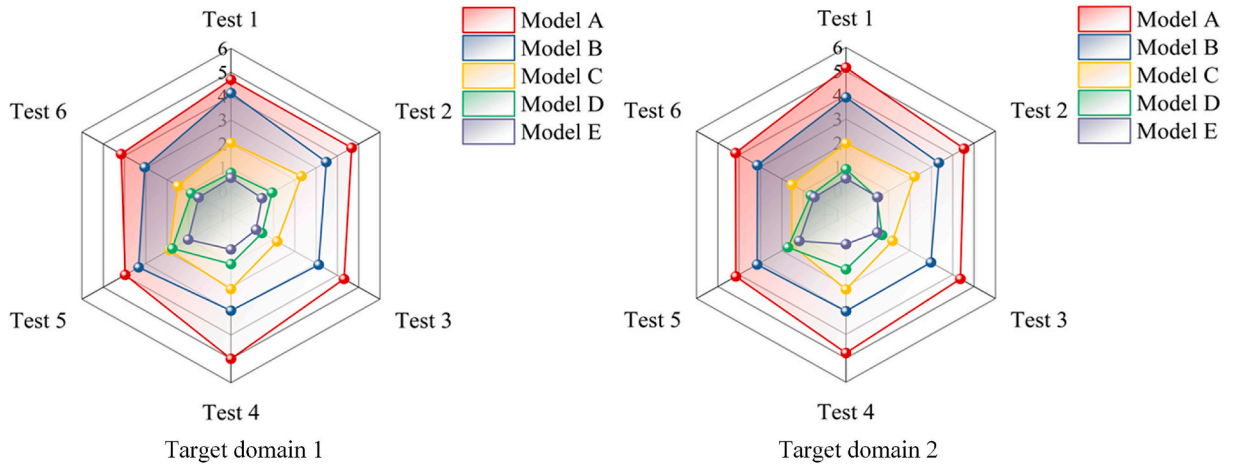


Fig. 18. Radar chart of accuracy standard deviation.

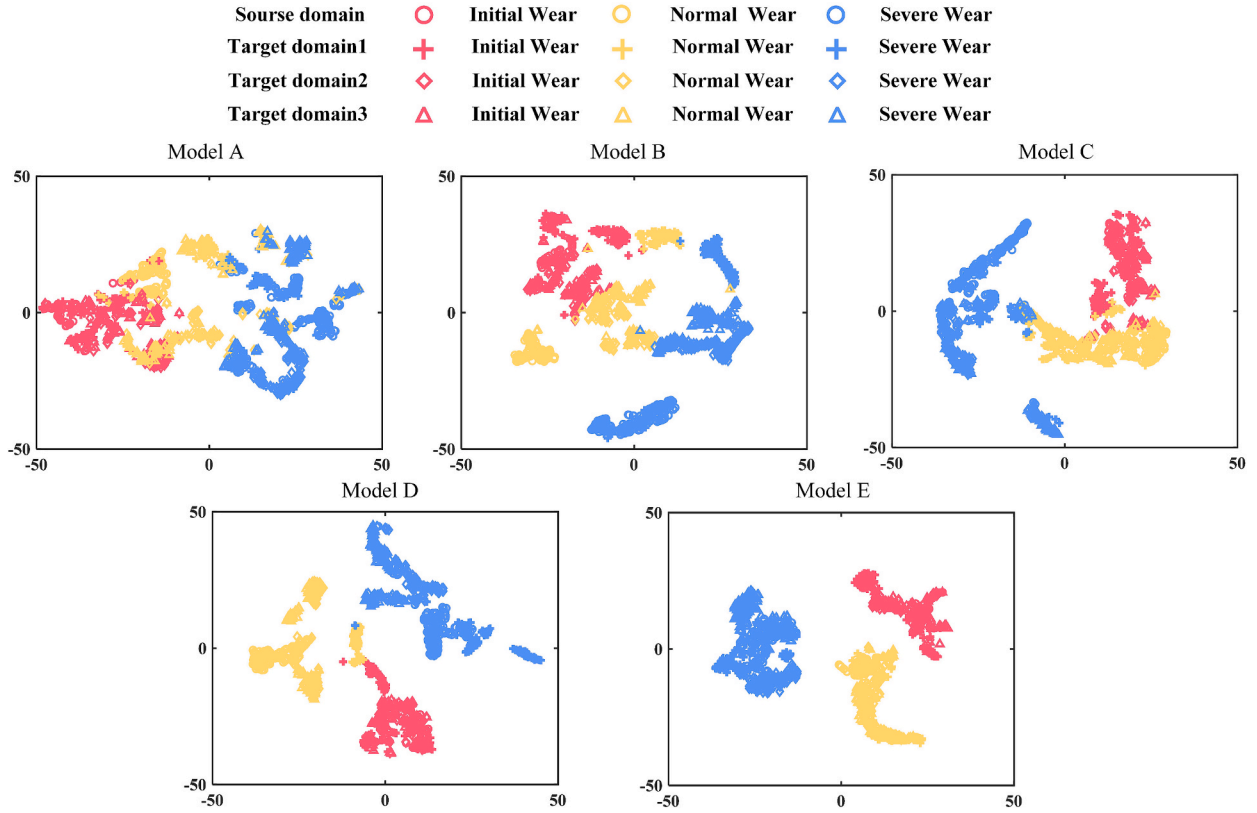


Fig. 19. t-SNE visualization features for transfer Test 7.

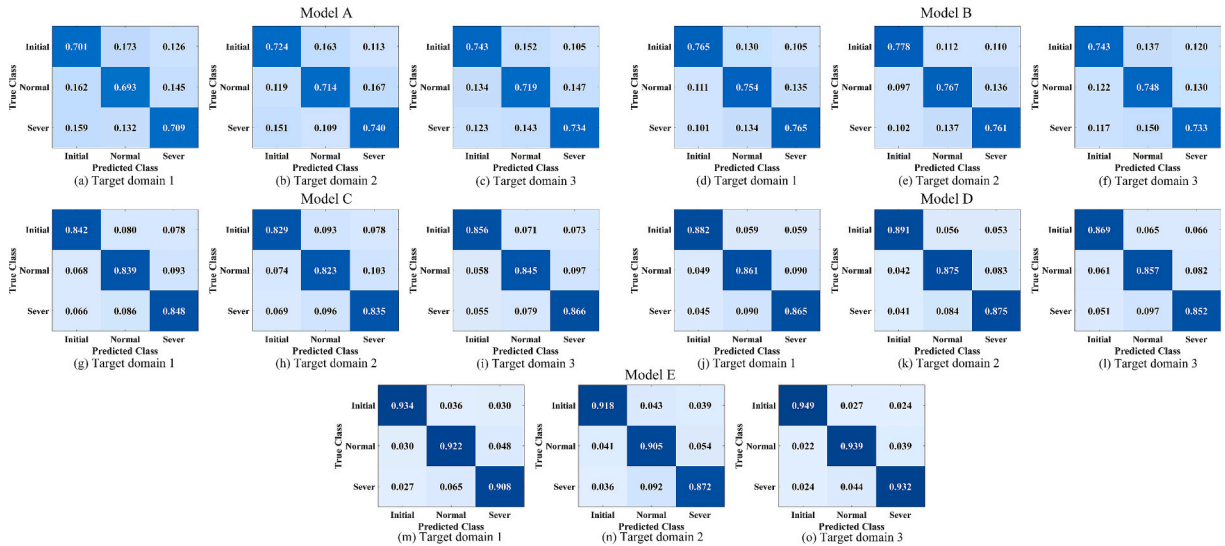


Fig. 20. Confusion matrix of multi-target domain samples in Test 7.

conditions. Model C improves the average accuracy to 84.26%, achieving more balanced performance across fault types and demonstrating reduced variability between working conditions, indicative of a degree of cross-domain robustness. Model D further enhances the average accuracy to 86.97% while maintaining strong diagnostic precision across all fault categories. The proposed Model E delivers the best overall performance, attaining accuracies of 92.13%, 89.83%, and 94.00% in the three target domains, with an average accuracy of 91.99%. These findings confirm the superior generalization and reliability of the proposed approach compared with existing methods. To assess the capability of the MCDADAN model in identifying tool wear states under real machining

conditions, a comprehensive evaluation was conducted using the GH4169 dataset. As shown in Fig. 21, the evaluation outcomes of five competing models are depicted as polar coordinate plots. Across various wear stages and target domains, the proposed model consistently achieved the highest performance, exhibiting clear superiority over baseline approaches. During the initial wear phase, the model demonstrated excellent classification accuracy, achieving an average recall of 0.934 and a precision of 0.940 across the three target domains. Under the more challenging normal wear condition, it maintained strong robustness, yielding a precision of 0.900 and an F1-score of 0.911. Even in the severe wear stage, the model sustained high stability, attaining a precision of 0.921. With its high classification accuracy and minimal misclassification rate, the proposed MCDADAN framework distinctly stands out, underscoring its practical applicability and deployment potential for tool wear monitoring in GH4169 high-temperature alloy machining. To further assess deployment feasibility, Table 9 reports the computational cost and inference latency of DANN and MCDADAN. It should be noted that the adversarial discriminator is employed only during offline alignment training; therefore, edge deployment executes only the prediction branch. Although DANN is slightly more efficient in terms of computational cost, the proposed MCDADAN introduces only a modest overhead while maintaining millisecond-level latency on the edge platform. Combined with the consistently improved recognition performance observed across the above multi-target transfer experiments, these results indicate that MCDADAN achieves a favorable balance between accuracy and efficiency, supporting its practical deployment in industrial shop-floor environments. To further strengthen the experimental comparisons under the multi-target transfer setting, we additionally implement three typical and competitive domain adaptation baselines, namely CDAN [54], AMDA [55], and AMEAN [56]. The quantitative results are summarized in Table 10. As reported in Table 10, MCDADAN delivers the best overall performance and consistently outperforms these advanced baselines across the multi-target transfer tasks. Overall, these results demonstrate that MCDADAN provides reliable and accurate tool wear recognition under varying machining conditions, particularly in the multi-target transfer setting.

5. Conclusion

This study proposed a novel machining condition decoupling adversarial domain adaptation network for multi-target domain tool

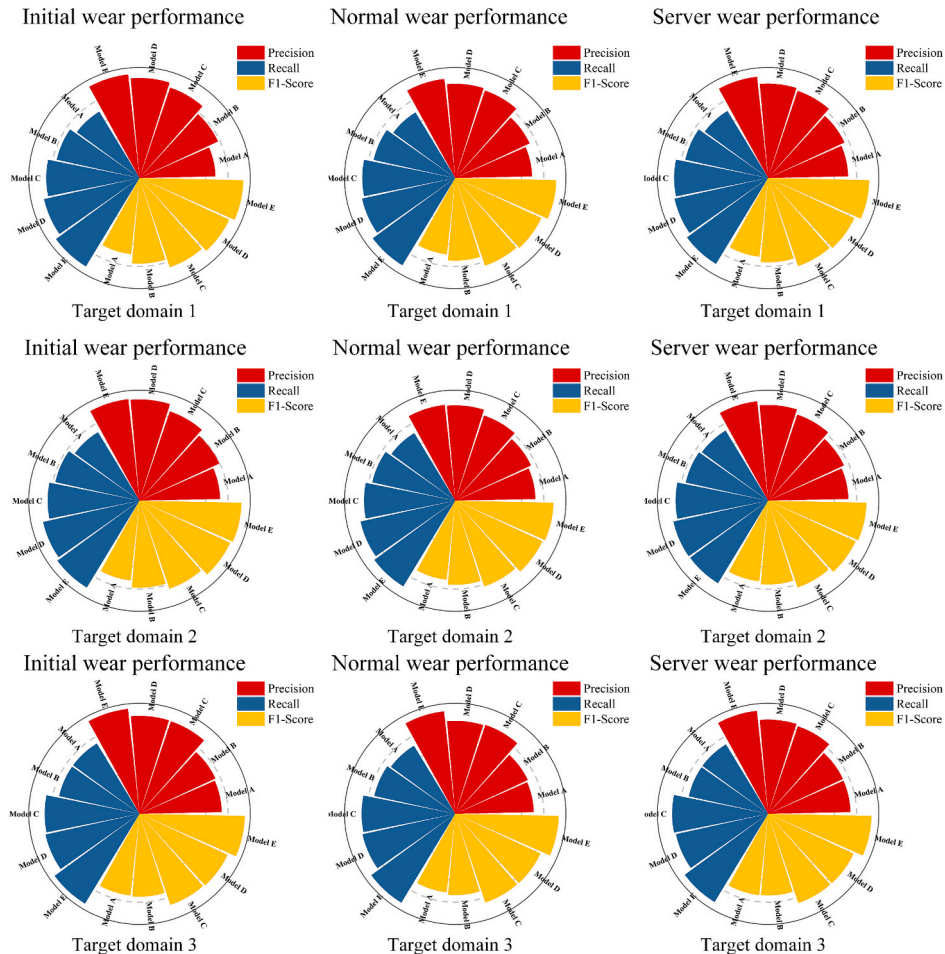


Fig. 21. Performance analysis of different models in Test 7.

Table 9
Computational complexity and inference latency analysis.

Model	Params (M)	FLOPs (G)	Latency (Server GPU)	Latency (Edge/Jetson)
DANN	1.25	0.015	1.2 ms	8.5 ms
MCDADAN	1.35	0.017	1.5 ms	9.8 ms

Table 10
Performance analysis of different models.

Source domain	Case 1	
Target domain	Case 2	Case 3
CDAN	79.48 $\pm$ 5.65	81.93 $\pm$ 4.22
AMDA	86.12 $\pm$ 3.04	85.42 $\pm$ 3.48
AMEAN	91.66 $\pm$ 2.13	89.19 $\pm$ 1.96
MCDADAN	97.31 $\pm$ 0.58	97.44 $\pm$ 0.52

wear state recognition, addressing generalization limitations arising from distributional discrepancies across diverse operating conditions in complex industrial environments. The following conclusions can be drawn:

- (1) Grounded in the generalization error bound theory of domain adaptation, this study proposes a blended alignment strategy tailored for multi-target domain scenarios. By integrating a shared wavelet convolutional feature extractor and a prototype discrepancy alignment mechanism, the approach effectively mitigates feature distribution disparities across domains, clarifies class specific prototype representations, and substantially enhances both cross-domain adaptability and feature discriminability of the model.
- (2) The proposed machining condition variation decoupling module leverages domain discriminator weights and multi-domain discrepancies to filter out operating condition-related interference thereby extracting more robust domain-invariant tool wear features.
- (3) A novel domain discrepancy loss formulation is introduced that combines an improved prototype boundary loss to align the source domain with multiple target domains and MMD based mechanism for mutual alignment among target domains. Multi-condition and multi-target transfer experiments on the GH4169 milling dataset and the PHM 2010 demonstrate that the proposed loss functions significantly improve diagnostic accuracy, reduce transfer bias, and effectively alleviate the issue of inconsistent domain matching in multi-domain adaptation tasks. Specifically, the model achieves an average diagnostic accuracy of 97.63% when two working conditions are used as the target domains, and 91.99% under a three-target-domain setting. Moreover, the precision, recall, and F1-score consistently outperform those of existing methods, verifying the superior generalization capability and stability of the proposed MCDADAN model.

At present, tool wear monitoring continues to be a major research focus within the machining domain. In future research, given that the statistical volume of available data may fall short of the training requirements for data-intensive models at the individual level, an important challenge is how to effectively train models with limited samples. Few-shot learning-based tool wear monitoring warrants more comprehensive investigation.

CRedit authorship contribution statement

Shunxiang Yao: Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shuoxue Sun:** Methodology, Investigation. **Tao Sun:** Writing – review & editing, Visualization. **Yuwen Sun:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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